A SCALED MAPPING APPROACH FOR TREATING SLOPING INTERFACES IN RANGE-DEPENDENT PARABOLIC EQUATION SOLUTIONS

by

Daniel Moran

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Golden, Colorado

Date _____

Signed: _____
Daniel Moran

Signed: _____
Dr. Jon M. Collis
Thesis Advisor

Golden, Colorado

Date _____

Signed: _____
Dr. Willy A. Hereman
Professor and Head
Department of Applied Mathematics and Statistics

ABSTRACT

An area of research in ocean acoustics is on solution techniques for environments with sloping seafloors. In these range-dependent environments the oceanic waveguide varies in the direction of propagation and exact analytic solutions do not exist. In order to enforce continuity conditions at the seafloor, the environment is approximated as a series of range-independent regions. An alternative to this approach, the mapping solution [M. D. Collins et al., JASA 107] applies a change of variables that results in a vertical translation of the environment and makes it easier to enforce ocean bottom interface conditions. An alternate mapping approach that uses a characteristic depth length scale is considered here. In contrast to the earlier mapping solution, the new approach uses a change of variables in which the depth variable is scaled relative to the depth of the bottom interface. The earlier approach distorted the surface, whereas the new mapping does not, which is a potential improvement. Along with the simplification of bottom interface conditions in both approaches, extra terms arising from the mapping are neglected. The new approach is implemented in a parabolic equation solution and is benchmarked against an existing solution. The accuracy of this new approach is established initially for problems involving fluid sediments.

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CHAPTER 1

INTRODUCTION

Acoustics is the study of wave propagation through media (generally fluid or elastic). Acoustic energy, or sound, can be categorized as waves of the compressional type, where energy is transferred as a result of compression and expansion of particles in the medium conducting the wave, whether that medium is the atmosphere, ocean, or solid earth, for example. Underwater acoustics is the study of sound in water. Sound propagation in the ocean is of interest in many fields of study including mathematics, physics, geology, geophysics, and biological studies. Sound is of such importance in the ocean because it can travel to great distances in this environment, whereas all forms of electromagnetic radiation are only able to travel comparably short distances. This proves useful for communications, navigation, underwater detection, and the studying of the structure of the seafloor. Therefore the study and modeling of such acoustic waves in the ocean environment is of great importance.

Sound speed in the ocean is an increasing function of temperature, salinity and pressure, and it characterizes sound transmission through the medium. These parameters themselves are influenced by factors including geographic location, time of day and season. Since ocean and other water environments consist of a layer of liquid between air and a more dense bottom, a natural waveguide is created because of the strong impedance jumps at the sea surface and seafloor. An important consideration in this field of study is reflection, scattering and transmission into the seafloor.

The formula used to describe the relationship between the angles of incidence and refraction of waves passing through an interface between two media is called Snell's law. The law states

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{c_1}{c_2} = \frac{n_2}{n_1} \tag{1.1}$$

where θ is the angle measured from the normal to the interface, c is the speed of sound in the medium, and n is the refractive index of the medium. This formula (which describes the

interaction of a wave at an interface, seen in Figure 1.1) along with interface conditions, is important when considering wave propagation in the oceanic waveguide.

As sound travels in the natural waveguide of the deep ocean, the sound is refracted within the ocean as a result of pressure and temperature variation (also governed by Snell's law), allowing it to transmit almost without loss. Energy loss occurs at the boundaries, especially in a lossy seafloor. In shallow waters, even more interesting interactions occur with steep seafloors, such as majority penetration, where almost no sound is reflected back into the water.

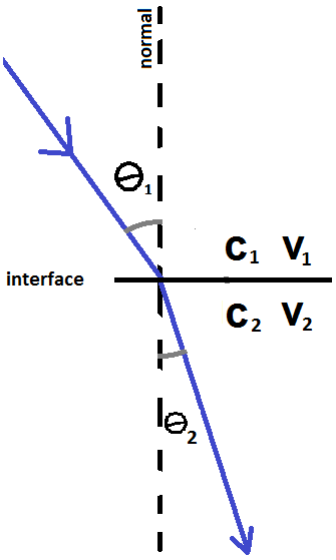


Figure 1.1: Refraction at the interface between two media with different refractive indices, n and wave speeds, c.

Various numerical methods to model underwater acoustics exist, and may be divided into two categories, either range-independent or range-dependent solutions. The traditional coordinate system used for these methods is a cylindrical coordinate system with the r-axis (range) along the interface and the z-axis (depth) passing through the source oriented downward. Range-independent models assume environmental parameters such as sound-speed, water depth, and bottom composition are invariant with range. When modeling more complex environments, for example an environment that is known to have variable bathymetry, a range-dependent solution is required. Range-dependent solutions allow that the ocean can

vary as a function of both depth and range. Note that surface and bottom reflection losses are incorporated via boundary conditions or reflection coefficients. Attenuation in the sediment can also be incorporated into range-dependent models. Attenuation is the reduction of the intensity of sound energy propagating in a medium, measured in amount of loss per length (for example decibels per wavelength). The ocean's surface is typically assumed to be horizontal (flat), but the ocean floor is often much more complexly shaped, and can have the property that it supports additional forms of waves (shear, elastic and associated interface waves) to propagate. Models often treat the bottom as another fluid layer with a higher density and higher sound speed than the water column. Models that treat the bottom as an elastic medium are not considered in this work.

This thesis considers range-dependent environments and uses a so-called parabolic equation solution for simulations. The parabolic equation (PE) method was introduced into the field of underwater acoustics in 1977 (Tappert, 1977). Parabolic equation techniques are based on factoring and approximating the operators in a wave equation to create one-way wave equations which govern incoming and outgoing energy. The PE method allows for medium changes at each range step and is computationally efficient because of the operator factorization allowing for a solution of essentially only half of the wave equation. The method has been adapted to solve either fluid or elastic media problems (the former treats compressional waves, and the latter treats both compressional and shear waves) and is both accurate and efficient.

The most important application of the PE method is to range-dependent problems, and hence this is still an active area of research. There are various methods to obtain a solution to the underwater acoustic problem using the parabolic approximation (discussed in Chapter 2.3). The PE method's initial numerical solution technique was based on the fast Fourier transform. This implementation is able to model deep-water problems well, but problems involving floor interaction can be inaccurate when considering wide-angle propagation, large sound-speed variations, and density discontinuities (such as at the ocean bottom interface).

Since then, the PE method has undergone significant improvements, including accuracy, usability (a term meaning the newer approaches are able to handle a wider range of problems), and efficiency. The technique vital to this work is the split-step Padé solution, which allows for very wide propagation angles because of its rational-linear approximation to differential operators in the wave equation. This technique is used in state-of-the-art PE calculations in underwater acoustics today due to its ability to solve the PE efficiently without losing accuracy, and to produce solutions for very wide angles of propagation. This solution technique also allows for large range steps, essential for a computationally efficient model, and is the "most efficient PE algorithm that has been developed" (Collins & King, 1995).

In order to handle range-dependent environments, PE solutions take advantage of the fact that in range-independent environments, the factorization of the wave equation is exact. Solutions use various techniques such as "stair-step" approximations and coordinate transformations (explored in detail in Chapter 2.4) to simplify the domain to exploit this fact for range-independent regions. One technique in particular that this thesis explores is a coordinate mapping technique for problems involving variable slopes analogous to a solution originally developed by Collins (Collins & Dacol, 2000).

The goal of this thesis is to use an alternate mapping approach, which has roots in various areas of mathematics, to handle range-dependent problems. This approach uses a scaled coordinate mapping to simplify the bottom interface without affecting the surface interface (whereas the original mapping does). The method outlined in this thesis is a modification to existing implementations which use stair-step approximations to handle range dependence. The new method's mapping essentially removes, via transformation, the range dependence of the problem, and then the solution is translated back into the original domain.

In Chapter 2, some preliminary mathematics are outlined. First, the acoustic wave equation is derived; then parabolic equation solutions are derived; numerical implementations are discussed along with various methods used to handle range dependence. Chapter 3 derives and discusses the new mapping approach this thesis is introducing. Preliminary results of

the solutions are shown and discussed in Chapter 4, and in Chapter 5, conclusions and future work are discussed.

CHAPTER 2

BACKGROUND

This chapter provides preliminary mathematics and specific details on the split-step Padé parabolic equation solution, which is the basis for most PE calculations performed in underwater acoustics, as well as techniques for handling range dependence.

2.1 Preliminary Mathematics

The wave equation governs the propagation of sound waves and hence is essential to understanding and modeling acoustic phenomena. To derive the wave equation, it is beneficial to consider what quantity is wanted to be determined, whether it is particle displacements, velocity, displacement potentials, or pressure. By considering the forces that tend to return a stretched string to its equilibrium position, a wave equation is derived. Consider a string of uniform linear density, ρ, and negligible stiffness, stretched to a tension T great enough so effects due to gravity can be ignored. Also assume no other external forces, such as friction. Figure 2.1 shows an infinitesimal element of the string with equilibrium position x and equilibrium length dx. If the transverse displacement of this element from its equilib-

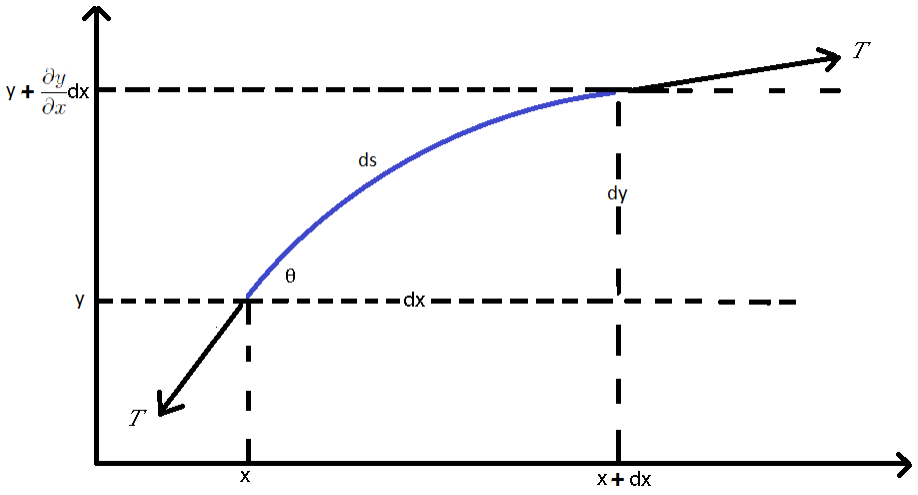


Figure 2.1: Segment of an elastic string under constant tension

rium position y is small, the tension T remains constant along the string and the difference between the y components of the tension at the two ends of the element is

$$F^y = (T\sin\theta)^{x+dx} - (T\sin\theta)^x, \tag{2.1}$$

where θ is the angle between the tangent to the string and the x axis, F^y indicates forces in the y-direction, $(T\sin\theta)^x$ is the value of $T\sin\theta$ at x, and dx is a small change in x. Applying the Taylor's series expansion

$$f(x+dx) = f(x) + \left(\frac{\partial f}{\partial x}\right)dx + \frac{1}{2}\left(\frac{\partial^2 f}{\partial x^2}\right)dx^2 + \ldots \tag{2.2}$$

to (2.1) gives

$$F^y = \left[(T\sin\theta) + \frac{\partial(T\sin\theta)}{\partial x}dx + \ldots\right] - (T\sin\theta) = \frac{\partial(T\sin\theta)}{\partial x}dx, \tag{2.3}$$

where only the lowest-order nonvanishing terms have been retained. If θ is small, $\sin\theta \approx \tan\theta = \partial y/\partial x$, and the net transverse force on the element becomes

$$F^y = \frac{\partial}{\partial x}\left(T\frac{\partial y}{\partial x}\right)dx = T\frac{\partial^2 y}{\partial x^2}dx. \tag{2.4}$$

Since the mass of the element is $\rho\ dx$ and its acceleration in the y direction is $\partial^2 y/\partial t^2$, Newton's second law gives

$$F^y = \rho dx\frac{\partial^2 y}{\partial t^2}. \tag{2.5}$$

Combining equations (2.4) and (2.5) yields a one-dimensional wave equation,

$$T\frac{\partial^2 y}{\partial x^2} = \rho\frac{\partial^2 y}{\partial t^2}. \tag{2.6}$$

Defining the wavespeed as $c = \sqrt{T/\rho}$ gives an equivalent equation,

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 y}{\partial t^2}, \tag{2.7}$$

which is the traditional form of the wave equation.

2.2 Derivation of the Fluid Parabolic Equation

This section describes derivations and solutions of parabolic equations for fluid media. By assuming that the solution to the wave equation in three dimensions is separable in space and time with a time-harmonic source and substituting the separable form of the solution into the wave equation results in what is referred to as the Helmholtz equation. To derive

the parabolic equation for fluid media, we begin with the Helmholtz equation for a constant-density medium, in cylindrical coordinates (r, θ, z), with a time-harmonic point source of time dependence $e^{-i\omega t}$:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + \frac{\partial^2 p}{\partial z^2} + \frac{1}{r^2}\frac{\partial^2 p}{\partial \theta^2} + k^2 p = 0, \tag{2.8}$$

and we assumed azimuthal symmetry, so there is no dependence on the θ coordinate. Here, r is the range, z is the depth, $p(r, \theta, z)$ the acoustic pressure, $k(r, \theta, z) = \omega/c(r, \theta, z)$ the medium wavenumber, ω the circular frequency, and c the sound speed. Ignoring azimuthal coupling gives $p = p(r, z)$, and reduces equation (2.8) to

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + \frac{\partial^2 p}{\partial z^2} + k^2 p = 0. \tag{2.9}$$

Substituting $\tilde{p} = \sqrt{r}p$ into equation (2.9) which removes the factor $r^{-1/2}$ from the complex pressure p, accounting for cylindrical spreading gives

$$\frac{\partial^2 \tilde{p}}{\partial r^2} + \frac{\partial^2 \tilde{p}}{\partial z^2} + k^2 \tilde{p} + \frac{1}{4r^2}\tilde{p} = 0. \tag{2.10}$$

Dropping tildes for simplicity, making the far-field assumption ($kr \gg 1$), and dropping $O(r^{-2})$ terms in equation (2.10), we obtain the elliptic wave equation:

$$\frac{\partial^2 p}{\partial r^2} + \frac{\partial^2 p}{\partial z^2} + k^2 p = 0. \tag{2.11}$$

Define the depth operator X as

$$X = \frac{1}{k_0^2}\left(\frac{\partial^2}{\partial z^2} + k^2 - k_0^2\right), \tag{2.12}$$

where $k_0 = \omega/c_0$ is the reference wavenumber for c_0, a representative sound speed. Note that the depth-dependence of the problem is entirely in the X operator. Hence we write the elliptic wave equation (2.11) as

$$\left(\frac{\partial^2}{\partial r^2} + k_0^2(1 + X)\right) p = 0. \tag{2.13}$$

Factoring equation (2.13) into two components (incoming and outgoing waves) gives

$$\left(\frac{\partial}{\partial r} - ik_0\sqrt{1+X}\right)\left(\frac{\partial}{\partial r} + ik_0\sqrt{1+X}\right) p - \left[\frac{\partial}{\partial r}, ik_0\sqrt{1+X}\right] p = 0, \tag{2.14}$$

where the bracketed quantity $[\Phi, \Psi] = \Phi\Psi - \Psi\Phi$ is the commutator of the operators Φ and Ψ. When considering range-independent problems, the two operators commute and the last

term in equation (2.14) is zero. For weak range dependence, the commutator is ignored (assumed small), and equation (2.14) becomes

$$\left(\frac{\partial}{\partial r} - ik_0\sqrt{1+X}\right)\left(\frac{\partial}{\partial r} + ik_0\sqrt{1+X}\right)p = 0. \tag{2.15}$$

Combined, both factors in equation (2.15) give incoming and outgoing wave equations. By assuming backscatter is negligible and retaining only the outgoing wave results in the fluid medium parabolic wave equation

$$\frac{\partial p}{\partial r} = ik_0\sqrt{1+X}p, \tag{2.16}$$

the formal solution to which is

$$p(r+\Delta r, z) = e^{ik_0\Delta r\sqrt{1+X}}p(r,z), \tag{2.17}$$

for range step Δr.

As noted, the approximation between equations (2.14) and (2.15) is exact for range-independent environments, and so this one-way wave equation is exact for these situations. To take advantage of this fact in range-dependent environments, the environments are often modeled as a sequence of range-independent regions, referred to as "stair-steps". This approximation will be discussed in Section 2.4.

2.2.1 Numerical Implementation of Depth Operator

In order to conserve energy when solving the parabolic equation numerically, the depth operator X is modified by using a modified dependent variable $\tilde{p} = p/\alpha$, where $\alpha = (\rho/k)^{1/2}$. The modified depth operator is of the form

$$\hat{X} = k_0^{-2}\left(\frac{\rho}{\alpha}\frac{\partial}{\partial z}\frac{1}{\rho}\frac{\partial}{\partial z}\alpha + k^2 - k_0^2\right). \tag{2.18}$$

This operator is discretized using Galerkin's method, described in (Collins & Westwood, 1991). This replaces the depth operator with a tridiagonal matrix, and this tridiagonal system of equations is then solved repeatedly.

2.3 Split-Step Parabolic Equation Solution Techniques

In numerical analysis, a split-step method is a numerical method used to solve nonlinear partial differential equations. The name arises because they use "splitting" or "fractional step" methods to solve variable coefficient equations (Tappert *et al.*, 2000).

The solution of the parabolic equation (2.16) or (2.17) requires a representation of the square root operator. Various methods of solving the one-way wave equation exist; we review those of implicit finite-difference and split-step Padé solutions.

The split-step Fourier solution was the basis for the first parabolic equation implementations in ocean acoustics. The solution is based on the fast Fourier transform, which allows for highly efficient solutions. However with this approach, only one assumption on the PE solution can be made (either energy propagates nearly horizontally, or speed of sound varies weakly). The result of this are accuracy limitations, and so there was a desire for another solution technique, which led to the development of a finite-difference Crank-Nicolson solution. The split-step Fourier solution remains in use because of its efficiency, but it is not often used for low frequency (<1 kHz) 2D problems since the introduction of the split-step Padé solution (Collins, 1993). The split-step Padé approach provides both highly accurate solutions, and comparable efficiency.

Both split-step solutions are based on approximating a function that involves an exponential square root operator. The split-step Fourier solution is based on a linear approximation of the square root operator and neglects the commutator. The approximation allows only a small number of horizontal wavenumbers, and by ignoring the commutator, the size of the range step must be about a wavelength (Küsel, 2005). Conversely, the split-step Padé solution has no restriction on horizontal wavenumber nor range step because it is based on a rational linear approximation of the exponential operator, and thus is the only split-step method described below.

2.3.1 Finite Differences

The parabolic wave equation is a one-way wave equation containing a square root of the depth operator X given in equation (2.16). In order to solve the partial differential equation numerically, an approximation must be made to the square root operator. Traditional approximations, based on Taylor series, give rise to the narrow-angle and wide-angle parabolic equations,

$$\frac{\partial p}{\partial r} = ik_0 \left(1 + \frac{1}{2}X\right) p, \tag{2.19}$$

and

$$\frac{\partial p}{\partial r} = ik_0 \left(\frac{1 + \frac{3}{4}X}{1 + \frac{1}{4}X}\right) p, \tag{2.20}$$

where p is the acoustic pressure at the points (r, z).

The narrow-angle parabolic equation (2.19) uses the first two terms in the Taylor series expansion of the square root function (Tappert, 1977):

$$\sqrt{1+X} = 1 + \frac{1}{2}X - \frac{1}{8}X^2 + \dots \qquad . \tag{2.21}$$

Using additional terms from the Taylor series is inefficient due to the fact that since X is raised to higher powers further approximations to higher-order derivatives would be necessary. The wide-angle parabolic equation (2.20) (Claerbout, 1976) uses a rational linear approximation involving first powers of X, a special case of the following Padé series (Collins, 1989):

$$\sqrt{1+X} - 1 = \sum_{j=1}^{n} \frac{a_{j,n}X}{1 + b_{j,n}X} + O(X^{2n+1}), \tag{2.22}$$

where n is the number of terms in the Padé expansion, and coefficients $a_{j,n}$ and $b_{j,n}$ are determined based on stability and accuracy constraints. These constraints can be specified by different methods, for example, by matching derivatives of the exponential function at $X = 0$ yielding

$$a_{j,n} = \frac{2}{2n+1} \sin^2\left(\frac{j\pi}{2n+1}\right), \tag{2.23}$$

and

$$b_{j,n} = \cos^2\left(\frac{j\pi}{2n+1}\right). \tag{2.24}$$

2.3.2 Narrow and Wide Angle Solutions

The solutions to equations (2.19) and (2.20) can be obtained by first discretizing the depth operators using Galerkin's method and then the resulting system is solved with a Crank-Nicolson finite-difference scheme in range. Finite-difference methods require step sizes which are relatively small compared to the acoustic wavelength, and are therefore less efficient for long-range propagation problems.

In order to determine whether or not the parabolic equation is narrow or wide angle, it depends on the approximation used for X. We define

$$n(z) = \frac{k(z)}{k_0} \tag{2.25}$$

as the index of refraction, which is a function of depth only in a homogeneous region. Then X can be rewritten as:

$$X \equiv \left(n^2 - 1 + \frac{1}{k_0^2}\frac{\partial^2}{\partial z^2}\right). \tag{2.26}$$

For a locally plane wave solution,

$$p = e^{\{i(k(r)r \pm k(z)z\}} \implies \frac{\partial^2}{\partial z^2} = -k(z)^2, \tag{2.27}$$

as seen in (2.17), and additionally it implies

$$\frac{1}{k_0^2}\frac{\partial^2}{\partial z^2} = -\frac{k(z)^2}{k_0^2}. \tag{2.28}$$

Note that the propagation angle θ with respect to the horizontal is defined by

$$\sin\theta = \pm\frac{k(z)}{k}. \tag{2.29}$$

Then from (2.25) and (2.29),

$$X = (n^2 - 1) - n^2\sin^2\theta = n^2\cos^2\theta - 1. \tag{2.30}$$

Using Snell's law, (1.1), X can be further reduced to

$$X = -\sin^2\theta_0. \tag{2.31}$$

A typical choice of c_0 is the sound speed at the source. Then X is only a function of source

radiation, so if X is small for propagation near the horizontal,

$$\sin\theta_0 \cong \pm\frac{k(z)}{k_0} = \frac{c(z)}{c_0}. \tag{2.32}$$

2.3.3 Split-Step Padé Solution

For the split-step Padé solution, the exponential square root operator in equation (2.17) is approximated by a product of rational linear functions, or higher-order Padé approximates

$$\exp\left(ik_0\Delta r\sqrt{1+X}\right) \approx \exp\left(ik_0\Delta r\right)\prod_{j=1}^{n}\frac{1+\alpha_{j,n}X}{1+\beta_{j,n}X}, \tag{2.33}$$

giving

$$p(r+\Delta r, z) = \exp\left(ik_0\Delta r\right)\prod_{j=1}^{n}\frac{1+\alpha_{j,n}X}{1+\beta_{j,n}X}p(r,z). \tag{2.34}$$

Coefficients $\alpha_{j,n}$ and $\beta_{j,n}$ are again determined based on stability and accuracy constraints. If the coefficients of the corresponding sum approximation are known, the coefficients of the product approximation can be specified. Combining the terms of the rational-linear sum into a common fraction gives

$$P(x)\prod_{j=1}^{M}\frac{1}{1+a_{2j,M}}, \tag{2.35}$$

where $P(x)$ is a polynomial. Finding the roots of the polynomial and factoring converts this representation into the product representation in equation (2.34) (Collins & Evans, 1992).

With enough terms in the rational approximation, the method allows for very wide-angle propagation, even for environments with large depth variations. It provides the accuracy of finite-difference solutions and the efficiency of split-step Fourier solutions. This solution has been the state-of-the-art solution since its introduction in 1993 and is the basis for most PE calculations performed in underwater acoustics today.

2.4 Range-Dependent Techniques

Coordinate transformation techniques are a common and very important part of many areas of mathematics. An example of this is a mapping solution in which a problem is translated into a new coordinate system to take advantage of the geometry of the new

system. For ocean acoustic problems, environmental parameters such as sound-speed, water depth and bottom composition often have some amount of range dependence. Therefore, there is a need to use various range-dependent techniques to simplify the environment in efforts to model the problem.

This thesis considers methods related to range-dependent bathymetry, which are motivated by strictly enforcing interface and boundary conditions. When solving underwater acoustic problems, the boundary conditions on the complex pressure are a pressure-release condition ($p = 0$) at the surface ($z = 0$). At the water-sediment interface at $z = d(r)$ both the pressure and normal component of particle velocity must be continuous. This work assumes the ocean bottom (sediment) is a fluid of differing sound speed and density. The next few subsections introduce the stair-step approximation and various coordinate transformations while highlighting the benefits of such techniques.

2.4.1 The Stair-Step Approximation

The stair-step discretization is the most commonly used technique for range-dependent problems. Range dependence in the form of a sloping ocean bottom is treated as a series of range-independent regions, resulting in a series of so-called "stair-steps" to approximate the bottom slope. These stair-steps approximate a slope with a series of horizontal and vertical regions. Since the factorization used to derive the PE is exact for range-independent regions, the PE is used to march the field through the region. Figure 2.2 shows a slope approximated by stair-steps, showing a few vertical segments separating media that have different acoustic properties. The interfaces are now horizontal and vertical along these stair-steps. Since the PE has two depth derivatives, both continuity of pressure and particle velocity can be explicitly enforced along each horizontal section. This is not the case along the vertical sections, however, due to the existence of only one range derivative. Often, especially in shallow environments, a correction must be applied to the vertical interfaces between regions (Porter *et al.*, 1991). This energy conservation correction states that for the

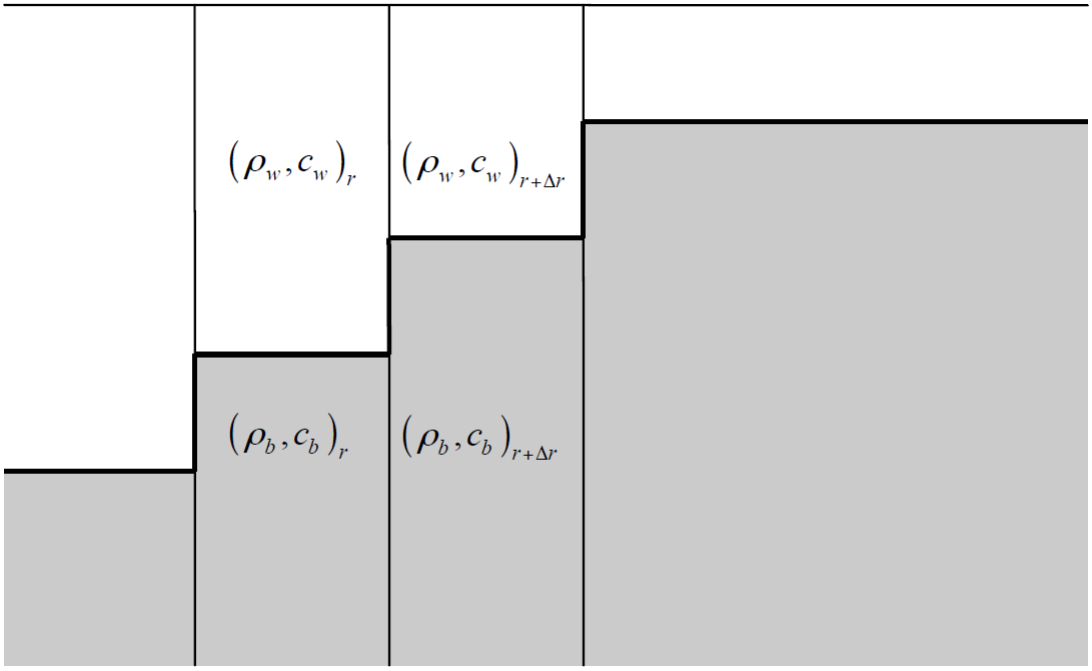


Figure 2.2: A slope is approximated by a series of "stair-steps".

energy E,

$$E(x) \approx \int \frac{k_0(x,z)\,|p|^2}{\rho} dz, \tag{2.36}$$

and a sufficient condition for conserving $E(x)$ is that the density reduced pressure $p/\sqrt{\rho c}$ is continuous. This energy conservation correction allows the PE solution to conserve energy flux across vertical interfaces.

2.4.2 Coordinate Transformations

A common mathematical transformation technique is a coordinate transformation from one system to another, such as from Cartesian to cylindrical coordinates. The geometry of a given problem may lend itself to one coordinate system over another. For example in acoustics, problems often involve a point source, or a line source, and the strength of the acoustic field is desired at various ranges. These problems lend themselves to cylindrical coordinates, with a fixed source position and the radius the distance from the source.

Another coordinate transformation technique takes advantage of boundary conditions of the problem. In underwater acoustics, the interface between water and air is taken to be a pressure-release boundary (pressure vanishes). This strong boundary condition is much simpler to enforce compared to the interface conditions at the water and the ocean floor as the pressure-release boundary condition is nondirectional (Collins & Siegmann, 1995). Thus if the geometry of the problem is complicated near the seafloor, a mapping technique may be used to simplify the geometry of the seafloor in favor of a more complicated surface geometry, but with an easier to enforce boundary condition.

Coordinate Mapping In underwater acoustics, coordinate mappings have been applied to problems involving variable slopes. At each range step, the medium is rigidly translated vertically to force the sloping interface into a horizontal interface (Outing, 2004).

The Mapping Approach The original approach of Collins (Collins & Dacol, 2000) is presented for the acoustic wave equation and is summarized here. Introduce the change of

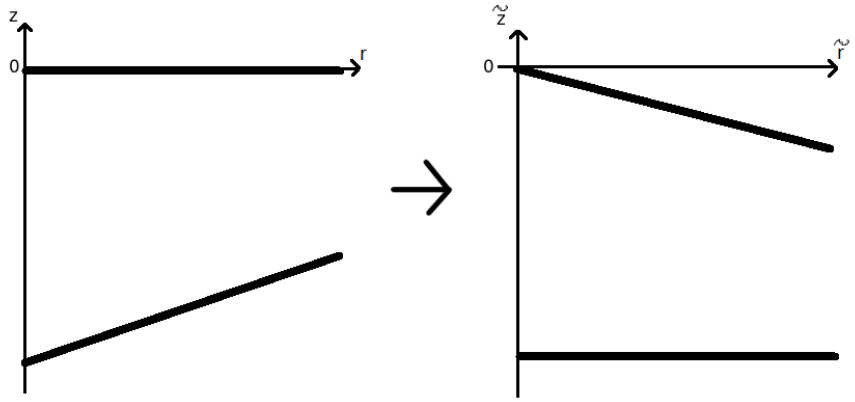


Figure 2.3: The original mapping solution's effect on the domain.

variables,

$$\begin{pmatrix} \tilde{r} \\ \tilde{z} \end{pmatrix} = \begin{pmatrix} r \\ z - d(r) \end{pmatrix} \tag{2.37}$$

for mapped coordinates, $\tilde{r}$ and $\tilde{z}$, and $d(r)$ the bottom depth at range r. This mapping, equation (2.37), replaces a sloping interface with a sloping upper boundary surface, (Fig-

ure 2.3), which is easier to handle. Assuming constant density, substituting into the far-field wave equation, (2.11), gives

$$\frac{\partial^2 \tilde{p}}{\partial \tilde{r}^2} - 2d' \frac{\partial^2 \tilde{p}}{\partial \tilde{r} \partial \tilde{z}} - d'' \frac{\partial \tilde{p}}{\partial \tilde{z}} + (d')^2 \frac{\partial^2 \tilde{p}}{\partial \tilde{z}^2} + \frac{\partial^2 \tilde{p}}{\partial \tilde{z}^2} + k^2 \tilde{p} = 0, \tag{2.38}$$

where $\tilde{p} = p(\tilde{r}, \tilde{z})$. Ignoring terms involving the slope (because we assume the slope to be small) gives

$$\frac{\partial^2 \tilde{p}}{\partial \tilde{r}^2} + \frac{\partial^2 \tilde{p}}{\partial \tilde{z}^2} + k^2 \tilde{p} = 0. \tag{2.39}$$

This equation can be factored using the parabolic approximation with only outgoing energy retained, giving the mapped parabolic equation

$$\frac{\partial \tilde{p}}{\partial \tilde{r}} = ik_0 \sqrt{1 + \tilde{X}} \tilde{p}, \tag{2.40}$$

where

$$\tilde{X} = \frac{1}{k_0^2} \left(\frac{\partial^2}{\partial \tilde{z}^2} + \tilde{k}^2 - k_0^2 \right) \tag{2.41}$$

and $\tilde{k} = k(\tilde{r}, \tilde{z}) = \omega / c(\tilde{r}, \tilde{z})$. Note that this is the exact form of the PE derived earlier, equation (2.16).

The mapping and neglect of the slope terms essentially results in physically distorting the waveguide, which conserves energy (Collins & Dacol, 2000). For problems with steep slopes, a correction must be applied to account for the neglected slope terms. This leading-order mapping correction, from equation (7) of (Collins & Dacol, 2000),

$$p(\tilde{r}, \tilde{z}) \to \exp\left(-ik_0 \tilde{z} \tan \delta\right) p(\tilde{r}, \tilde{z}) \tag{2.42}$$

is only applied at a point with a change in slope δ. Since the mapping solution avoids problems with vertical interfaces, the correction only affects the phase.

The next subsection describes the rotated parabolic equation solution, which is related to the mapping solution. This rotated solution is advantageous in that it does not give rise to extra terms in the wave equation. The rotated PE solution is much more complicated when working with problems involving variable slopes, whereas the mapping solution more naturally handles these.

The Rotated Parabolic Equation Solution The rotated parabolic equation solution involves rotating the coordinate axes so that it aligns with the bottom interface and then marches the PE solution parallel to the sloping seafloor. As before, the ocean surface becomes a sloping interface, and is treated by a sequence of stair-steps. New cylindrical coordinates are introduced through the rotation matrix so that

$$\begin{pmatrix} \hat{r} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} r \\ z \end{pmatrix} \tag{2.43}$$

where ϕ is the angle of rotation (angle of the bottom slope from horizontal), and $\hat{r}$ and $\hat{z}$ are tangent and normal to the ocean floor interface. Substituting into the far-field wave equation (2.11) yields

$$\begin{aligned} \frac{\partial^2 \hat{p}}{\partial \hat{r}^2}\cos^2\phi - 2\frac{\partial^2 \hat{p}}{\partial \hat{r}\partial \hat{z}}\cos\phi\sin\phi + \frac{\partial^2 \hat{p}}{\partial \hat{z}^2}\sin^2\phi + \frac{\partial^2 \hat{p}}{\partial \hat{r}^2}\sin^2\phi \\ + 2\frac{\partial^2 \hat{p}}{\partial \hat{r}\partial \hat{z}}\cos\phi\sin\phi + \frac{\partial^2 \hat{p}}{\partial \hat{z}^2}\cos^2\phi + k^2\hat{p} = 0, \end{aligned} \tag{2.44}$$

which simplifies to

$$\frac{\partial^2 \hat{p}}{\partial \hat{r}^2} + \frac{\partial^2 \hat{p}}{\partial \hat{z}^2} + k^2\hat{p} = 0, \tag{2.45}$$

with $\hat{p} = p(\hat{r}, \hat{z})$. This equation is factored, and only the outgoing wave is considered, resulting in the rotated parabolic equation

$$\frac{\partial \hat{p}}{\partial \hat{r}} = ik_0\sqrt{1 + \hat{X}}\hat{p}, \tag{2.46}$$

where

$$\hat{X} = \frac{1}{k_0^2}\left(\frac{\partial^2}{\partial \hat{z}^2} + \hat{k}^2 - k_0^2\right) \tag{2.47}$$

and $\hat{k} = k(\hat{r}, \hat{z}) = \omega/c(\hat{r}, \hat{z})$.

CHAPTER 3

DISCUSSION AND ANALYSIS

An alternative mapping solution is proposed using the existing mapping parabolic equation solution, (Collins & Dacol, 2000), discussed in the previous chapter as a reference. The form of the new mapping solution will be compared to the existing solution to reveal similarities and differences.

3.1 Alternate Mapping Solution

Consider the mapping $\tilde{z} = \frac{z}{d(r)}$ (as opposed to the original map $\tilde{z} = z - d(r)$). This new mapping is advantageous because the complicated geometry of the seafloor is mapped to a horizontal line without the surface geometry changing. There is no trade-off in geometry (favoring a complicated surface over a complicated seafloor), only a simplification of the seafloor geometry. The following analysis is carried out in a fashion similar to the previous mapping derivation in an effort to compare the different mapping approaches.

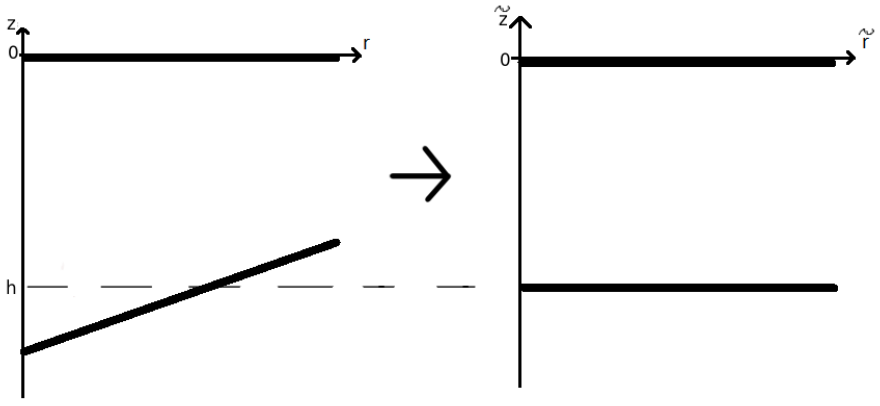


Figure 3.1: The new mapping solution's translation of the domain.

First, consider the depth of the seafloor $z_d = hd(r)$, where h a constant representative depth (shown in Figure 3.1) and $d(r)$ the relative nondimensional bottom depth at range r. Introduce the change of variables,

$$\begin{pmatrix} \tilde{r} \\ \tilde{z} \end{pmatrix} = \begin{pmatrix} r \\ \frac{z}{d(r)} \end{pmatrix}, \tag{3.1}$$

for mapped coordinates, $\tilde{r}$ and $\tilde{z}$. This mapping, equation (3.1), replaces a sloping interface with a horizontal interface at the representative depth h, and maintains the horizontal orientation of the upper boundary surface (seen in Figure 3.1). At both interfaces it is easier to apply boundary conditions explicitly. Changing differential operators based on the change of variables and substituting into the far-field wave equation (2.11), gives

$$\frac{\partial^2 \tilde{p}}{\partial \tilde{r}^2} - \frac{2z_d}{d^2} d' \frac{\partial^2 \tilde{p}}{\partial \tilde{r} \partial \tilde{z}} - \underbrace{\frac{z_d}{d^2} d'' \frac{\partial \tilde{p}}{\partial \tilde{z}}}_{\text{A}} + \underbrace{\frac{2z_d}{d^3} (d')^2 \frac{\partial \tilde{p}}{\partial \tilde{z}}}_{\text{B}} + \underbrace{\frac{z_d^2}{d^4} (d')^2 \frac{\partial^2 \tilde{p}}{\partial \tilde{z}^2}}_{\text{C}} + \underbrace{\frac{1}{d^2} \frac{\partial^2 \tilde{p}}{\partial \tilde{z}^2}}_{\text{D}} + k^2 \tilde{p} = 0, \tag{3.2}$$

where $\tilde{p} = p(\tilde{r}, \tilde{z})$. Terms involving only depth derivatives and slope terms have been labeled A, B, C, and D and will be discussed later. Ignoring terms involving the slope (assumed to be small, to be demonstrated later) gives

$$\frac{\partial^2 \tilde{p}}{\partial \tilde{r}^2} + \frac{1}{d^2} \frac{\partial^2 \tilde{p}}{\partial \tilde{z}^2} + k^2 \tilde{p} = 0. \tag{3.3}$$

This equation is again factored using the parabolic approximation and only outgoing energy is considered. This results in the new mapped parabolic equation

$$\frac{\partial \tilde{p}}{\partial \tilde{r}} = ik_0 \sqrt{1 + \tilde{X}} \tilde{p}, \tag{3.4}$$

where

$$\tilde{X} = \frac{1}{k_0^2} \left(\frac{1}{d^2} \frac{\partial^2}{\partial \tilde{z}^2} + \tilde{k}^2 - k_0^2 \right) \tag{3.5}$$

and $\tilde{k} = k(\tilde{r}, \tilde{z}) = \omega / c(\tilde{r}, \tilde{z})$.

This new mapping solution is nearly the same as the original mapping solution, but with an extra $\frac{1}{d^2}$ term multiplying the second partial derivative of $\tilde{p}$ with respect to $\tilde{z}$ (3.3). In the new mapping, additional terms that arise in the wave equation are neglected. In an effort to quantify that the neglected terms are in fact negligible, the values of terms B, C and D from (3.2) over range are plotted with respect to each other in Figure 3.2. The values for the terms were calculated based on the 0.23° slope seafloor example, discussed in Chapter 4 Section 1. The value of the A term for all examples discussed in this thesis is zero because

the slope of the seafloor is constant. The values of the terms are plotted on a logarithmic scale, which shows the three terms are orders of magnitude different from each other. The D term, which is the term we included in our derivation of the mapped parabolic equation, has the greatest value throughout the domain, justifying our neglect of the other terms. Analogous results were found for all other slopes presented in this thesis.

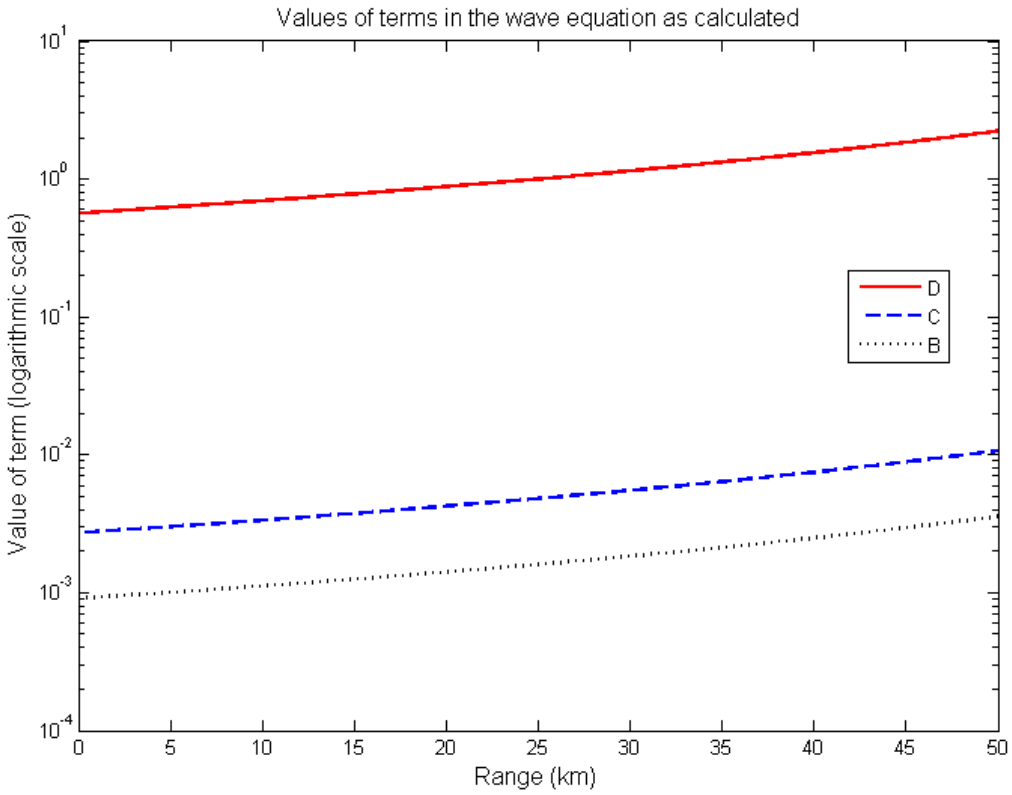


Figure 3.2: Comparing values of the terms in the wave equation.

Similar to the original mapping, this neglect of slope terms is equivalent to physically distorting the waveguide. Due to the similarities in results from the two mapping solutions, some correction must be applied to the new mapping solution to account for the effects of the neglected terms for problems with a steep bottom slope, in a manner similar to what is presented in (Collins & Dacol, 2000). Additionally, the new mapping must account for domain adjustment by transforming the mapped solution back into the original domain.

Currently, the mapping is being applied to the far field wave equation, (2.11), and then the parabolic approximation is introduced, but an interesting idea to consider would be to apply the mapping first to the Helmholtz equation and then introduce a parabolic approximation. For consistency with the previous mapping, (Collins & Dacol, 2000) this other option was not considered. However, because there is an exact relation between the solutions of the Helmholtz and parabolic equations for arbitrary two-dimensional waveguides (DeSanto, 1977), this other option could be investigated further outside of this work.

3.2 Domain Adjustment

The main effect that needs to be accounted for as a result of the alternate mapping is the distortion of the domain. The region in which the seafloor lies above the representative h must be expanded, and the region where the seafloor is below h must be compacted. This means that any depths used in calculations, $z(r)$, (such as source and receiver depth) must be adjusted to the mapped domain. This is accomplished by

$$\hat{z}(r) = z(r)/d(r) \tag{3.6}$$

where $\hat{z}$ is the new depth in the mapped domain.

After the solution is obtained, it must be mapped back into the original domain. To map the solution back into the original domain, the mapped solution at each range is interpolated into the original domain. At each range step, the maximum depth of the calculation routine must be adjusted to a corresponding maximum depth for the original domain. If the depth of the seafloor at the current range is less than h, the maximum depth is increased; if the seafloor depth is greater than h, the maximum depth is decreased. Accordingly, the grid spacing must either be increased or decreased for the new domain. By correcting the grid spacing, the values calculated by the mapped solution are adjusted into the original domain. After the interpolation has been completed across the entire domain, the alternate mapping solution has been obtained.

3.3 Variable Bathymetry Environments

For problems of variable bathymetry, the solution currently acts no differently than problems with only one constant slope. The sloped region between flat regions of the domain is used to calculate h, and the environment is mapped accordingly. It may be advantageous while expanding the solution to break the domain along the range into regions for each different slope. For each new sloped region, the mapping would be applied as described above, then the regions would be "stitched" together to ensure energy conservation across the regions.

CHAPTER 4

RESULTS AND ANALYSIS

A comparison of the results of the new mapping solution to a reference solution, RAM (Collins, 1995), will be conducted on various problems in order to reveal the strengths and limitations of the new mapping solution.

Solutions are presented in terms of transmission loss, defined by:

$$\mathrm{TL} = -20 \log \left| \frac{p(r,z)}{p_0} \right|, \tag{4.1}$$

where p is the acoustic pressure and p_0 is a reference pressure one meter from the source. Solutions are discussed via two types of plots. In each sequence of plots presented, three contour plots represent the following: the first contour plot shows the reference solution, the second plot shows the solution in the mapped domain, and the third plot shows the final solution after mapping and domain adjustment. The three contour plots, which show the transmission loss at each range and depth are followed by a line plot, showing the transmission loss at each range at a fixed depth, the dashed line indicating the reference solution, and the solid line indicating the new mapping solution.

The theoretical slope limitation on parabolic equation models is 7° (Jensen *et al.*, n.d.). As such, the examples presented below stay below this theoretical limit, and the solutions themselves are investigated to determine when the mapping solution breaks down.

4.1 Range-Independent Example

The acoustic and environmental parameters for this example is defined as follows: there is a 50 Hz source at $z_s = 50$ m; the sound speed in the water column is 1480 m/s. In the sediment the sound speed is 1700 m/s, the density is 1.5 g/cm^3, and the attenuation is 0.5 dB/λ. The bathymetry in this example is 400 m from $r = 0$ km to $r = 50$ km. The transmission loss plot is given at a receiver depth of $z_r = 50$ m.

In Figure 4.1, the new mapping solution (solid) and the reference solution (dashed) match throughout the range. This indicates the mapping solution is exact for range-independent problems.

4.2 Upslope Examples

The acoustic and environmental parameters in this section are defined as follows: there is a 50 Hz source at $z_s = 50$ m; the sound speed in the water column is 1480 m/s. In the sediment the sound speed is 1700 m/s, the density is 1.5 g/cm^3, and the attenuation is 0.5 dB/λ. The slope of the seafloor is positive for each example, but the slope and depth vary. Transmission loss plots are given at a receiver depth of $z_r = 50$ m.

4.2.1 0.23° Slope Seafloor

The bathymetry in this example is 400 m at $r = 0$ km and slopes linearly to 200 m at $r = 50$ km. The slope of the seafloor in this example is 0.23°. The reference solution is shown in Figure 4.2(a).

The mapping solution, Figure 4.2(c), shows similar energy propagation to the reference solution Figure 4.2(a), including penetration into the sediment. In Figure 4.2(d), the new mapping solution (solid) and the reference solution (dashed) closely follow each other with fine details of the curves in agreement. Error resulting from the new mapping is greatest between 5 and 15 km, and again between 20 and 30 km. The mapping solution shows about 1 less dB of transmission loss than the reference solution through this region, but the details of the curve are in agreement. The mapping solution matches very well with the reference solution.

4.2.2 0.46° Slope Seafloor

The bathymetry in this example is 500 m at $r = 0$ km, and slopes linearly to 100 m at $r = 50$ km. The slope of the seafloor in this example is 0.46°. The reference solution is

shown in Figure 4.3(a).

The mapping solution, Figure 4.3(c), shows similar energy propagation to the reference solution Figure 4.3(a), including penetration into the sediment. In Figure 4.3(d), the the new mapping solution (solid) and the reference solution (dashed) closely follow each other with fine details of the curves in agreement. Error resulting from the new mapping is greatest between 5 and 20 km, and again between 45 and 50 km. In the first region is the mapping solution shows about 3 less dB of transmission loss than the reference solution, and in the second region, the mapping solution shows about 3 more dB of transmission loss than the reference solution. Throughout the range, the details of the curve are in agreement, and the mapping solution matches very well with the reference solution.

4.2.3 2.3° Slope Seafloor

The bathymetry in this example is 500 m at $r = 0$ km, and slopes linearly to 100 m at $r = 10$ km. The slope of the seafloor in this example is 2.3°. The reference solution is shown in Figure 4.4(a).

The mapping solution, Figure 4.4(c), shows similar energy propagation to the reference solution Figure 4.4(a), including similar penetration into the sediment. In Figure 4.4(d), the new mapping solution (solid) and the reference solution (dashed) closely follow each other with fine details of the curve in agreement. Error resulting from the new mapping is greatest between 2 and 6 km, where the mapping solution shows about 2 less dB of transmission loss than the reference solution. From 0 to 2 km and 6 to 10 km, the details of the curve are in great agreement, and the mapping solution matches very well with the reference solution.

4.2.4 3.1° Slope Seafloor

The bathymetry in this example is 650 m at $r = 0$ km, and slopes linearly to 100 m at $r = 10$ km. The slope of the seafloor in this example is 3.1°. The reference solution is shown in Figure 4.5(a).

The mapping solution, Figure 4.5(c), shows similar energy propagation to the reference solution Figure 4.5(a) above the seafloor, but below the seafloor the differences are evident. Less energy penetrates the seafloor near the end of the region, after 7 km. In Figure 4.5(d), the new mapping solution (solid) and the reference solution (dashed) follow each other closely throughout the range. Error resulting from the new mapping is greatest between 2 and 3 km, where the mapping solution shows 5 less dB of transmission loss than the reference solution, and again at the end of the range. The mapping solution shows agreement with the fine details of the reference solution.

4.2.5 4.0° Slope Seafloor

The bathymetry in this example is 800 m at $r = 0$ km, and slopes linearly to 100 m at $r = 10$ km. The slope of the seafloor in this example is 4.0°. The reference solution is shown in Figure 4.6(a).

The mapping solution, Figure 4.6(c), is not as distinguished as the reference solution, Figure 4.6(a). In Figure 4.6(d), the new mapping solution (solid) and the reference solution (dashed) shows the most error, now between 2 and 3 km. The mapping solution shows nearly 5 less dB of transmission loss than the reference solution around in this range. Throughout the range, the curves are in agreement, and the mapping solution matches well with the reference solution.

4.3 Downslope Examples

The acoustic and environmental parameters in this section are defined as follows: there is a 50 Hz source at $z_s = 50$ m; the sound speed in the water column is 1480 m/s. In the sediment the sound speed is 1700 m/s, the density is 1.5 g/cm^3, and the attenuation is 0.5 dB/λ. The slope of the seafloor is negative for each example, but the slope and depth vary. Transmission loss plots are given at a receiver depth of $z_r = 50$ m.

4.3.1 $-0.23°$ Slope Seafloor

The bathymetry in this example is 200 m at $r = 0$ km, and slopes linearly to 400 m at $r = 50$ km. The slope of the seafloor in this example is $-0.23°$. The reference solution is shown in Figure 4.7(a).

The mapping solution, Figure 4.7(c), shows similar energy propagation to the reference solution Figure 4.7(a), including penetration into the sediment. In Figure 4.7(d), the new mapping solution (solid) and the reference solution (dashed) closely follow each other with fine details of the curves in agreement, albeit with nearly constant error. Throughout the range, the mapping solution shows about 2 more dB of transmission loss than the reference solution through this region, but the details of the curve are in agreement. The mapping solution matches very well with the reference solution.

4.3.2 $-0.46°$ Slope Seafloor

The bathymetry in this example is 100 m at $r = 0$ km, and slopes linearly to 500 m at $r = 50$ km. The slope of the seafloor in this example is $-0.46°$. The reference solution is shown in Figure 4.8(a).

The mapping solution, Figure 4.8(c), shows similar energy propagation to the reference solution Figure 4.8(a), including penetration into the sediment. In Figure 4.8(d), the new mapping solution (solid) and the reference solution (dashed) follow each other, but show major amplitude difference, even though the slope is small. Between the ranges of 0 and 20 km, the new mapping solution shows as an additional 10 dB in transmission loss versus the reference solution, but this discrepancy decreases to about 3 dB at 50 km.

4.3.3 $-2.3°$ Slope Seafloor

The bathymetry in this example is 100 m at $r = 0$ km, and slopes linearly to 500 m at $r = 10$ km. The slope of the seafloor in this example is $-2.3°$. The reference solution is shown in Figure 4.9(a).

The mapping solution, Figure 4.9(c), shows fairly similar energy propagation as the reference solution Figure 4.9(a), but penetration into the sediment is different. In Figure 4.9(d), the new mapping solution (solid) and the reference solution (dashed) follow each other, but again show major difference in loss. Between the ranges of 0 and 6 km, the new mapping solution shows an additional 10 dB in transmission loss versus the reference solution, but this discrepancy decreases to about 3 dB at 7 km. Despite this difference in loss, the new mapping solution matches the reference solution throughout the peaks and valleys of the plot.

4.3.4 -3.1° Slope Seafloor

The bathymetry in this example is 100 m at $r = 0$ km, and slopes linearly to 650 m at $r = 10$ km. The slope of the seafloor in this example is -3.1°. The reference solution is shown in Figure 4.10(a).

Comparing the mapping solution, Figure 4.10(c) to the reference solution, Figure 4.10(a), less energy penetrates the seafloor at the beginning of the region between 0 and 2 km. In Figure 4.10(d), the new mapping solution (solid) and the reference solution (dashed) follow each other, but show large difference in loss. Between the ranges of 3 and 4 km, the new mapping solution shows a much larger increase in the loss than the reference solution. Despite this large difference in loss, the new mapping solution matches the reference solution throughout the peaks and valleys of the plot.

4.3.5 -4.0° Slope Seafloor

The bathymetry in this example is 100 m at $r = 0$ km, and slopes linearly to 800 m at $r = 10$ km. The slope of the seafloor in this example is -4.0°. The reference solution is shown in Figure 4.11(a).

Again, comparing the mapping solution, Figure 4.11(c) and the reference solution, Figure 4.11(a), less energy penetrates the seafloor at the beginning of the region between 0 and

2 km. The transmission loss error shown in Figure 4.11(d) is again evident similar to the previous examples. The new mapping solution (solid) and the reference solution (dashed) follow each other, but throughout the range, the new mapping solution shows an additional 10 dB in transmission loss versus the reference solution. Despite this difference in loss, the new mapping solution matches the details of the reference solution.

4.4 Variable Bathymetry

Variable bathymetry is much more complicated than the example environments previously introduced. The acoustic and environmental parameters in this section are defined as follows: there is a 50 Hz source at $z_s = 50$ m; the sound speed in the water column is 1480 m/s. In the sediment the sound speed is 1700 m/s, density is 1.5 g/cm^3, and the attenuation is 0.5 dB/λ. The slope of the seafloor is flat for 0 to 20 km, positive for 20 to 30 km, then flat for 30 to 50 km, but the slope of the seafloor in the middle region varies, as does the depth. Transmission loss plots are given at a receiver depth of $z_r = 50$ m.

4.4.1 1.1° Variable Slope Seafloor

The bathymetry in this example is 400 m at $r < 20$ km, and slopes linearly to 200 m for $r > 30$ km. The slope of the seafloor in this example is 1.1°. The reference solution is shown in Figure 4.12(a).

The mapping solution, Figure 4.12(c), shows similar energy propagation to the reference solution Figure 4.12(a), including penetration into the sediment. In Figure 4.12(d), the new mapping solution (solid) and the reference solution (dashed) closely follow each other with fine details of the curves in agreement. Error resulting from the new mapping is greatest between 10 and 20 km, where the mapping solution shows about 3 less dB of transmission loss than the reference solution. Throughout the range, the details of the curve are in agreement, and the mapping solution matches very well with the reference solution. After 30 km, the new mapping solution shows more dB of transmission loss than the reference solution, but

the curves are still in great agreement.

4.4.2 2.3° Variable Slope Seafloor

The bathymetry in this example is 500 m at $r < 20$ km, and slopes linearly to 100 m at $r > 30$ km. The slope of the seafloor in this example is 2.3°. The reference solution is shown in Figure 4.13(a).

The mapping solution, Figure 4.13(c), shows similar energy propagation to the reference solution Figure 4.12(a), including penetration into the sediment until after the sloped portion of the seafloor. In Figure 4.13(d), the new mapping solution (solid) and the reference solution (dashed) closely follow each other with fine details of the curves in agreement until after the sloped portion of the seafloor. Error resulting from the new mapping is greatest at the end of the range, 50 km, where the mapping solution shows nearly 20 more dB of transmission loss than the reference solution. From 0 to 30 km, the new mapping solution consistently shows about 5 less dB of transmission loss, but after 30 km, the new mapping solution shows an increased rate of transmission loss compared to the reference solution.

4.4.3 Causes of Error

The first thought on the source of the error in the solution is the neglect of the additional terms in the mapped wave equation. To test this theory, the additional terms were calculated at each range step and plotted along side each other in Figure 3.2. The neglected terms are orders of magnitude smaller than the included term. The test environment used in the calculation of the additional terms is the same as the 0.23° slope environment introduced in the previous subsection. This plot illustrates that the additional terms truly are negligible (at least for the environments discussed here), and the source of the error must be somewhere else.

Another possible cause of error include lack of full energy conservation correction, and perhaps the lack of a complete mapping correction. In literature, the need for a complete

mapping correction is normally only necessary for problems with steep slopes, which may be why the later plots break down.

In an effort to better quantify the angle this new mapping solution breaks down at, an error quantification has been performed. In Figure 4.14(a) and Figure 4.14(b), the 2 norm of the error has been plotted versus the angle of the seafloor for various examples. There is no real marked increase in magnitude of error up to 4 degrees, however, and the error stays fairly constant even with the increase in error.

4.5 Summary

A new mapping approach for solving problems involving sloping interfaces in fluid sediments has been developed. By applying the mapping described by the change of variables shown in (3.1), the PE method can efficiently solve problems involving depth and range dependence while maintaining simple geometries for both the seafloor and surface. At each range step, the medium is mapped vertically either up or down to a fixed seafloor depth h. To simplify the approach, the slope is assumed small (less than 5°) and extra terms that arise in the wave equation due to the mapping are neglected.

Several examples demonstrate the capability of the method, benchmarked against a reference solution. Error is very low for upslope cases, but for downslope problems, the error is greater.

The new mapped parabolic equation has been applied to problems involving simple geometry. The new mapping solution eliminates the complex boundary condition of the original mapping solution, and similarly eliminates problems that arise from using stair-steps.

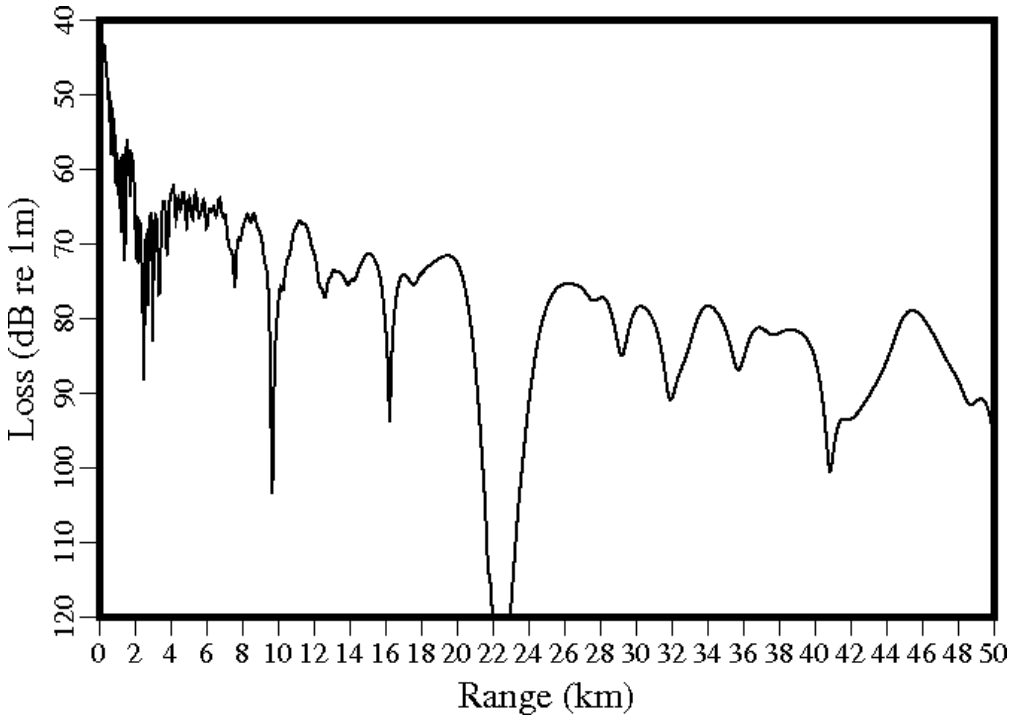
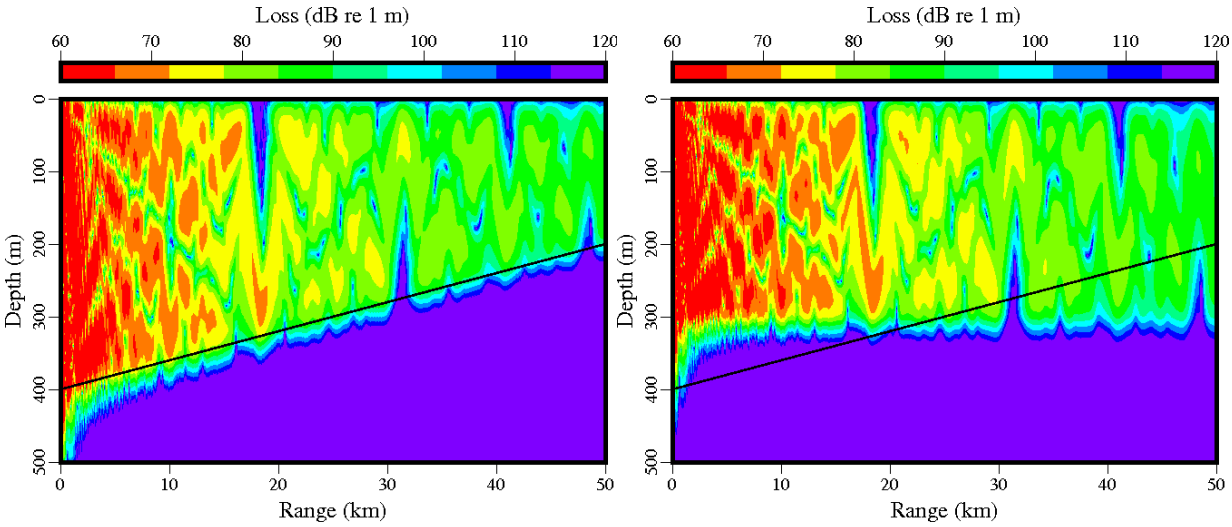
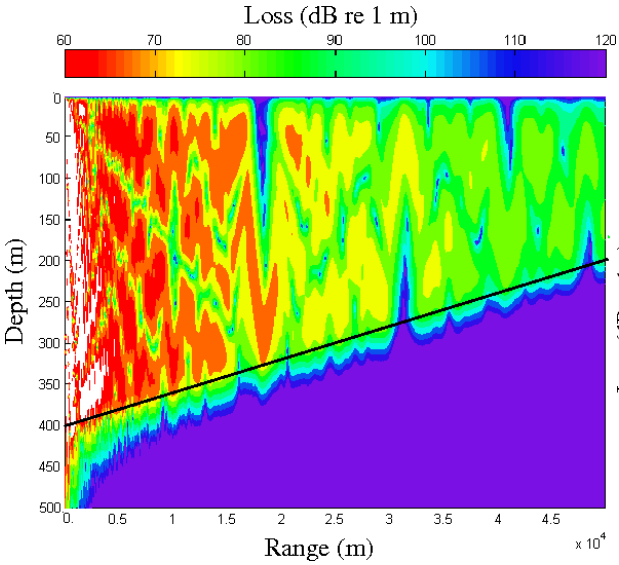


Figure 4.1: Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM).

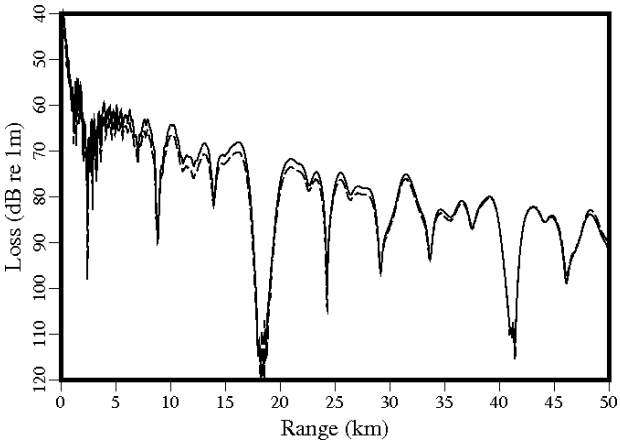


(a) The reference solution.

(b) The solution in the mapped domain.

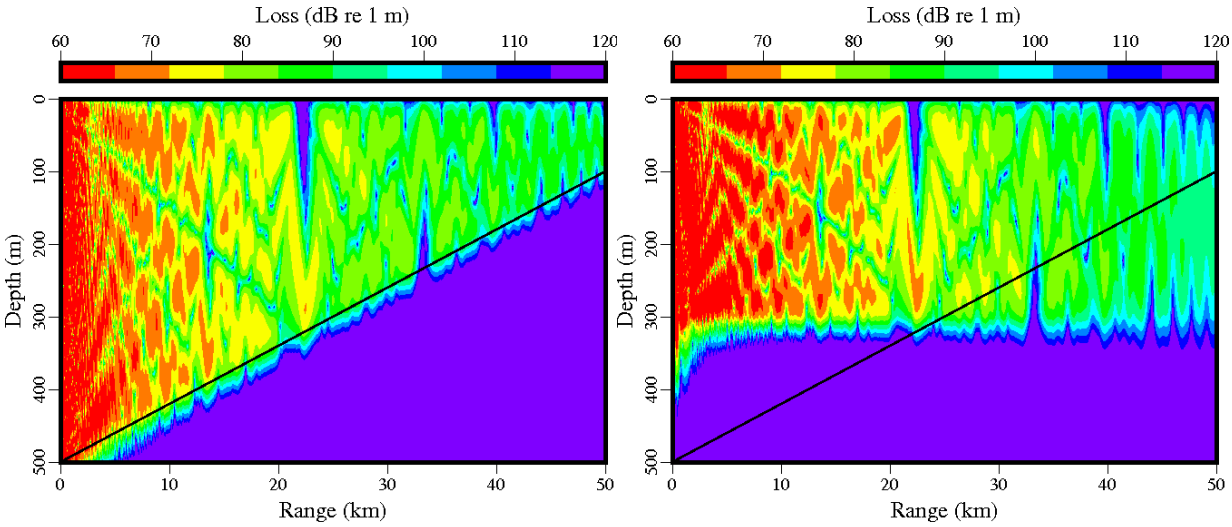


(c) The solution after mapping and domain adjustment have been applied.



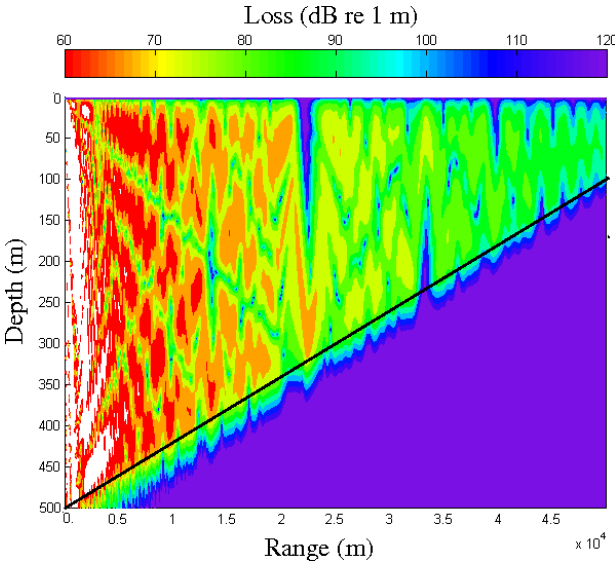
(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM).

Figure 4.2: A 0.23° bottom slope domain.

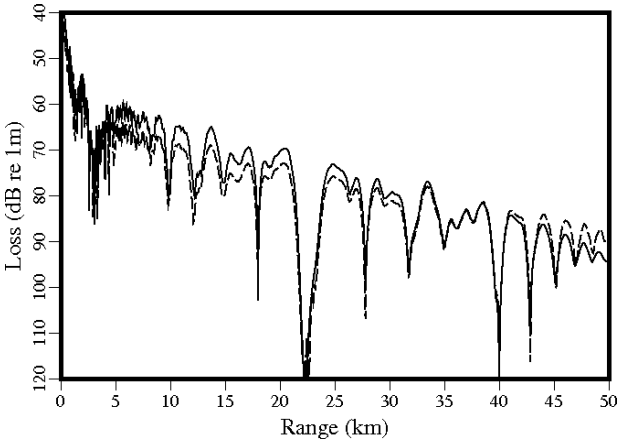


(a) The reference solution.

(b) The solution in the mapped domain.

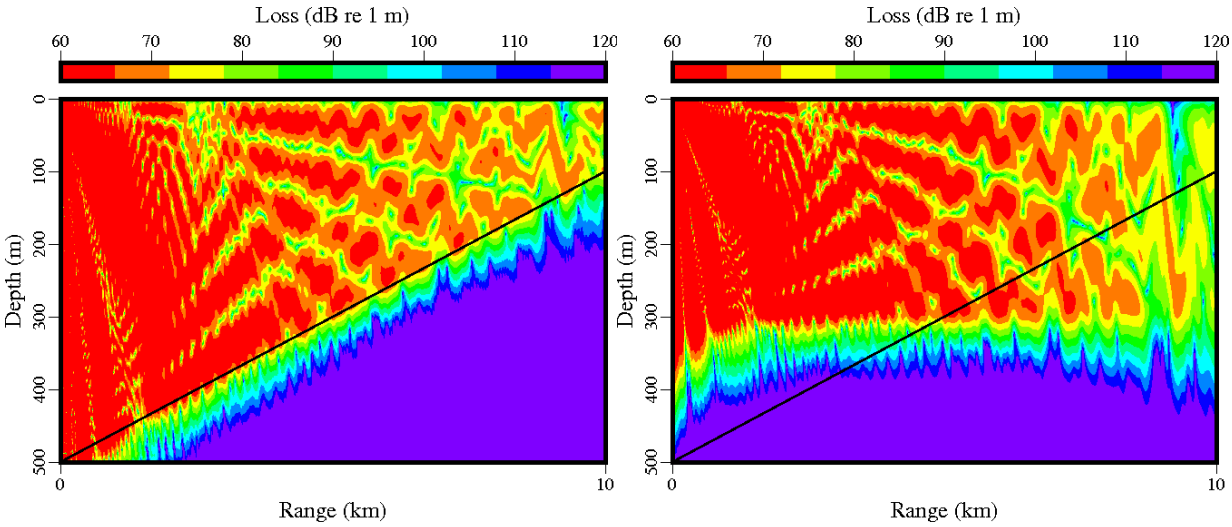


(c) The solution after mapping and domain adjustment have been applied.



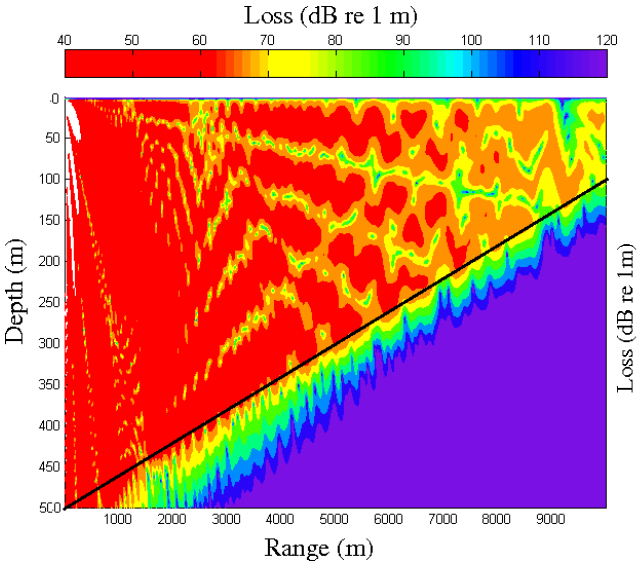
(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.3: A 0.46° bottom slope domain.

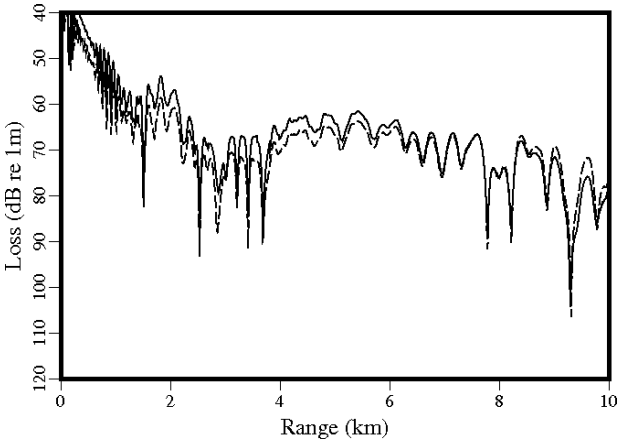


(a) The reference solution.

(b) The solution in the mapped domain.

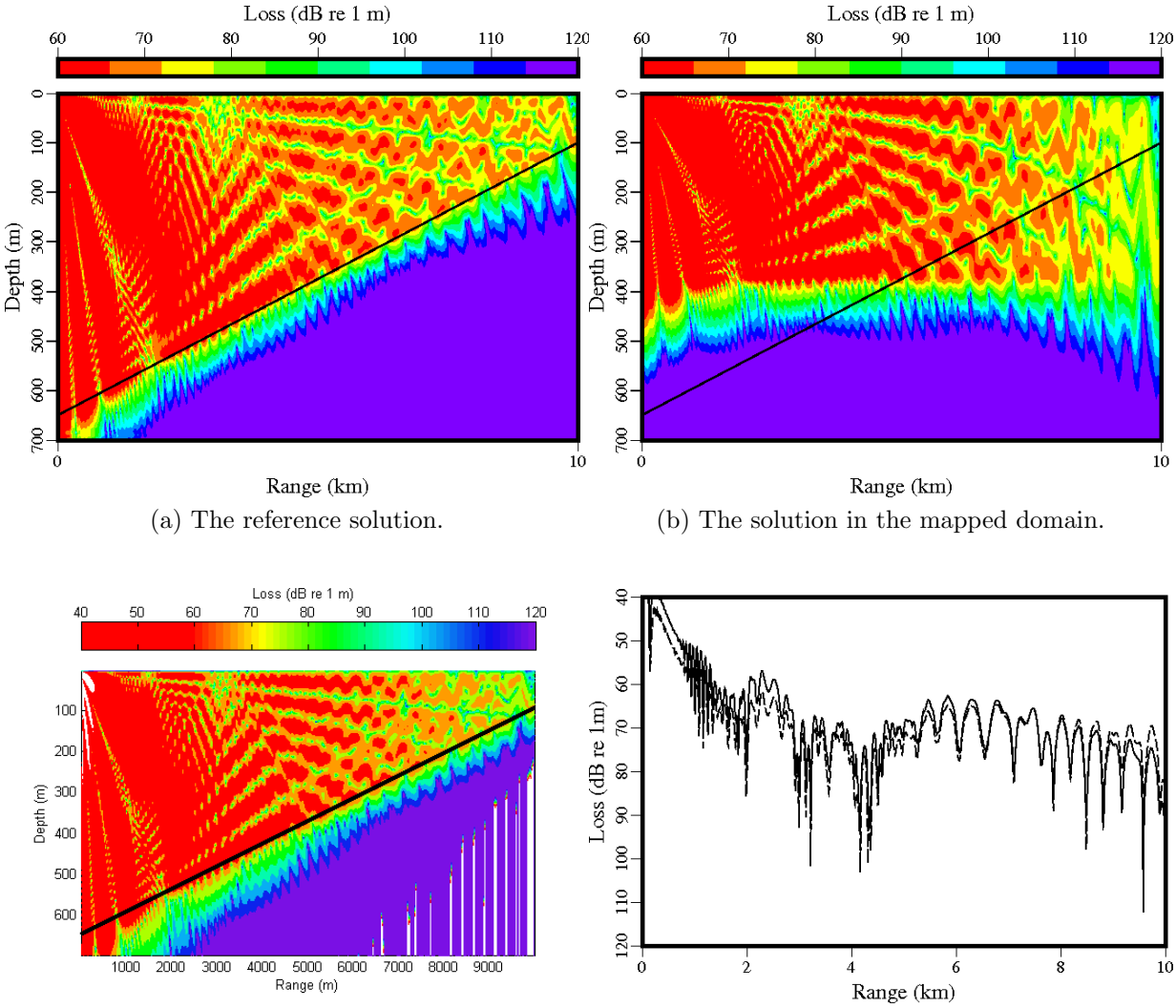


(c) The solution after mapping and domain adjustment hav been applied.



(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.4: A 2.3° bottom slope domain.



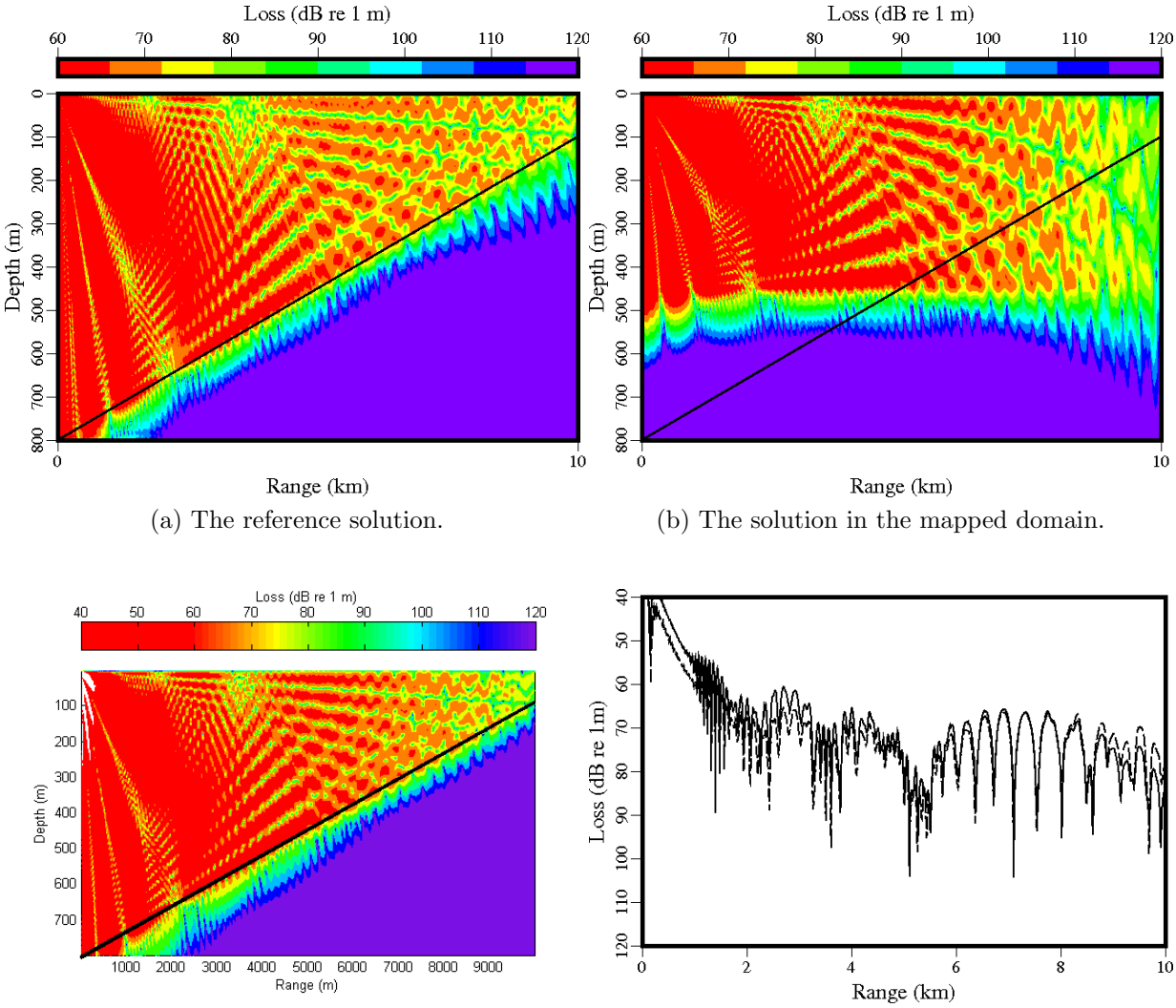
(a) The reference solution.

(b) The solution in the mapped domain.

(c) The solution after mapping and domain adjustment have been applied.

(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.5: A 3.1° bottom slope domain.



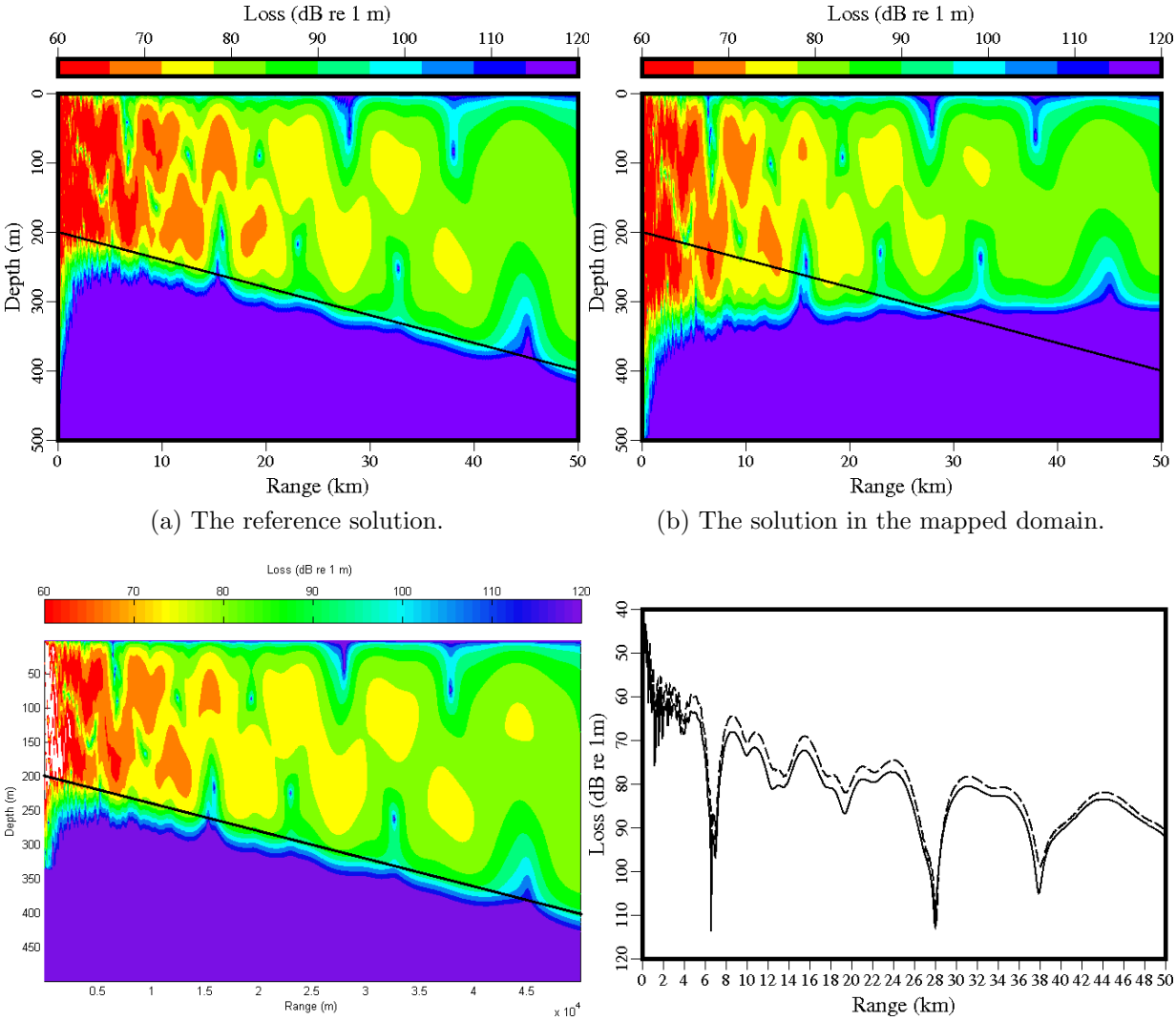
(a) The reference solution.

(b) The solution in the mapped domain.

(c) The solution after mapping and domain adjustment have been applied.

(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.6: A 4.0° bottom slope domain.



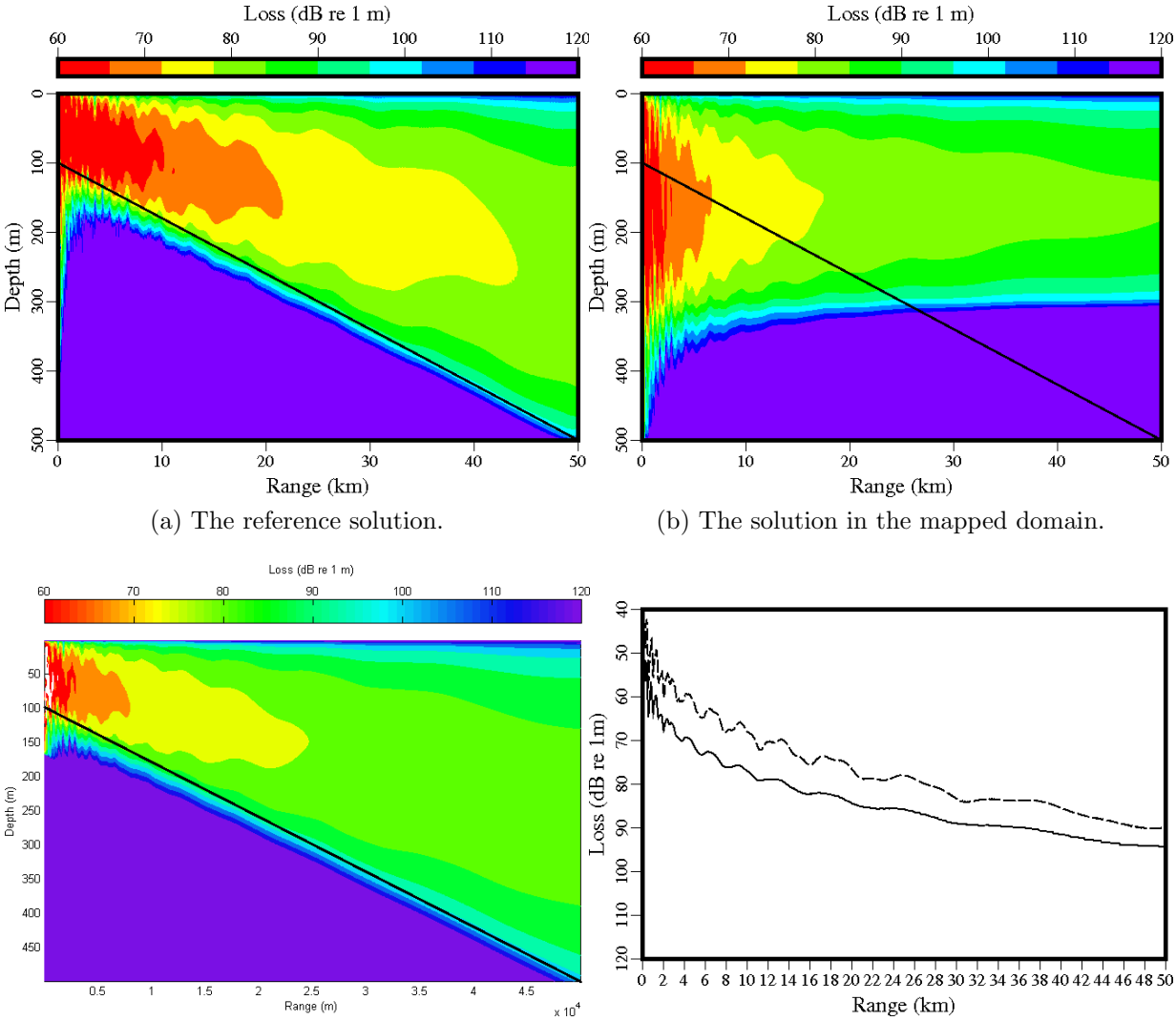
(a) The reference solution.

(b) The solution in the mapped domain.

(c) The solution after mapping and domain adjustment have been applied.

(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.7: A $-0.23°$ bottom slope domain.



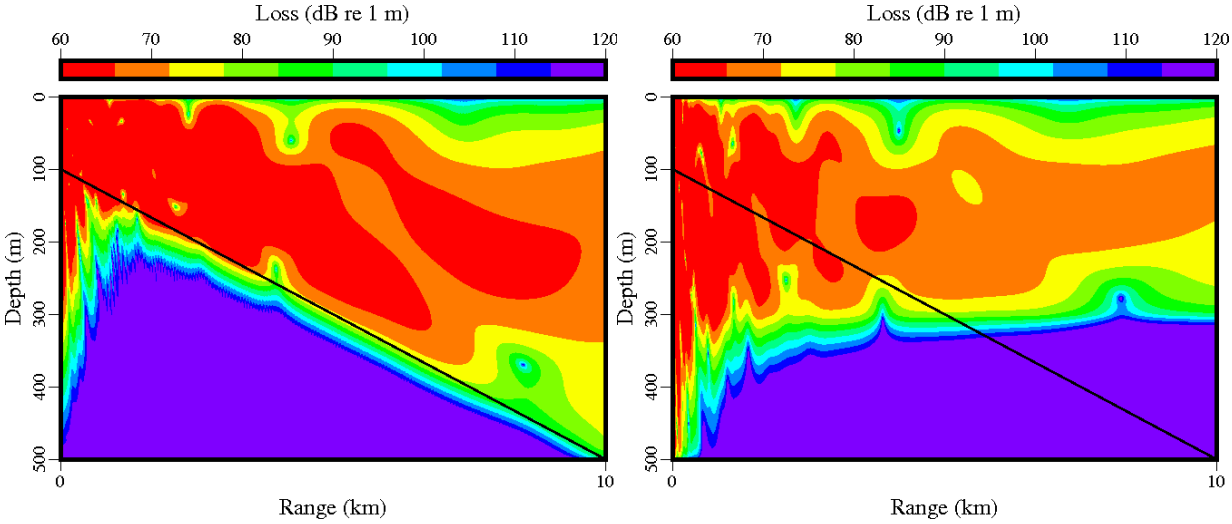
(a) The reference solution.

(b) The solution in the mapped domain.

(c) The solution after mapping and domain adjustment have been applied.

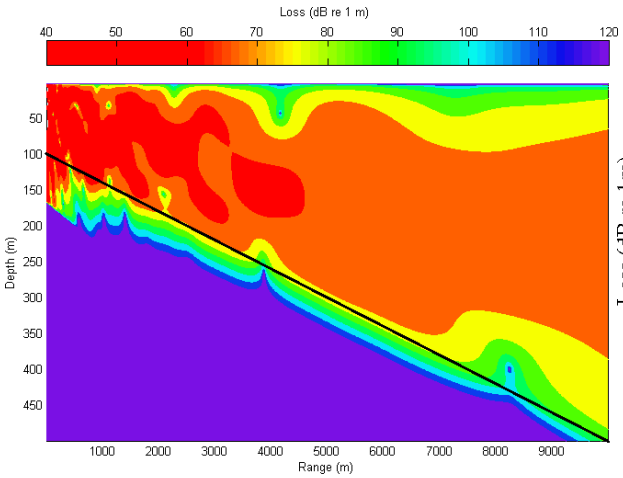
(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.8: A $-0.46°$ bottom slope domain.

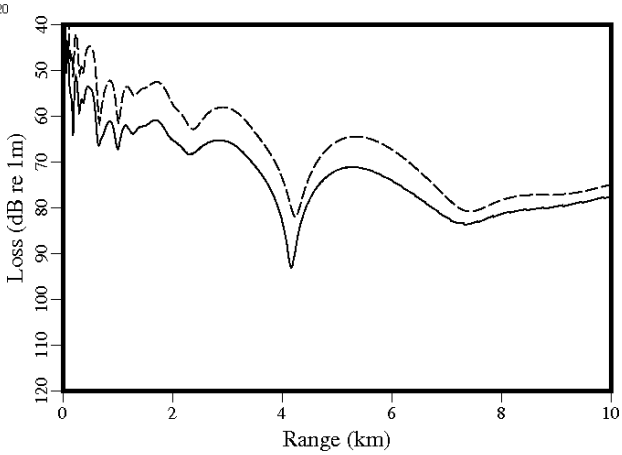


(a) The reference solution.

(b) The solution in the mapped domain.

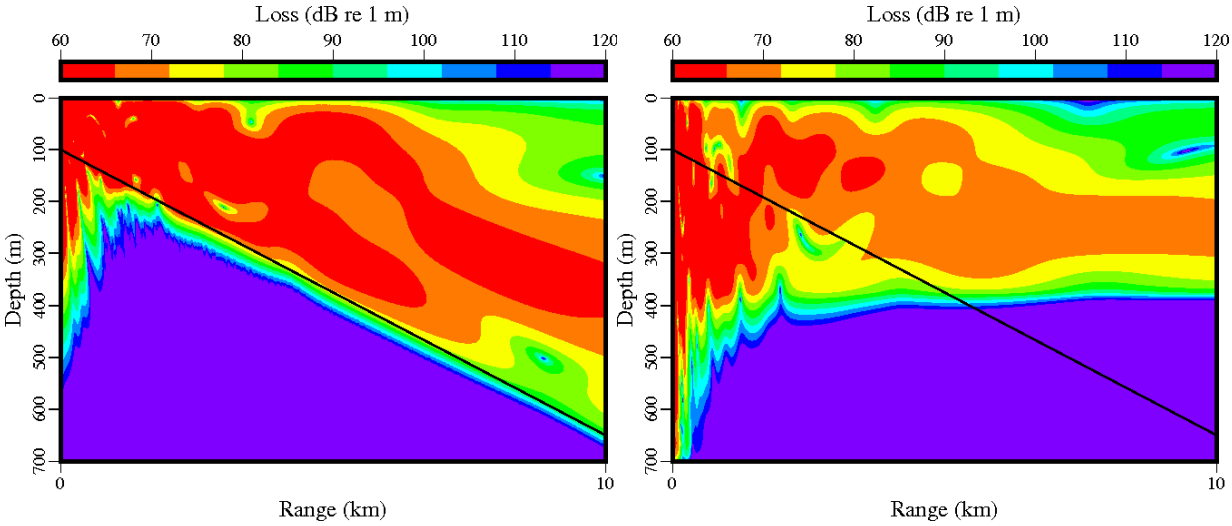


(c) The solution after mapping and domain adjustment have been applied.



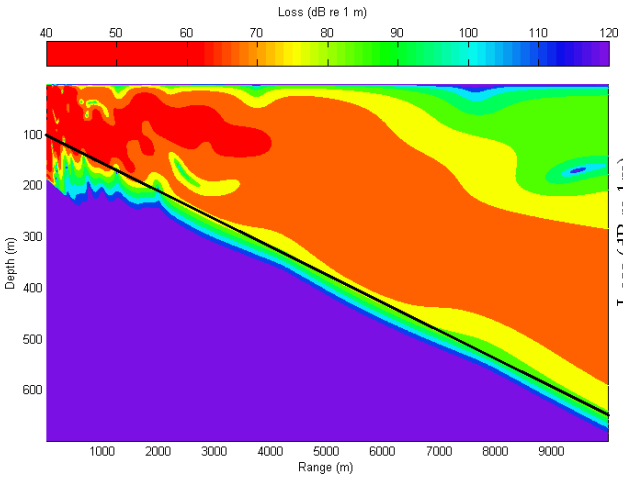
(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.9: A $-2.3°$ bottom slope domain.

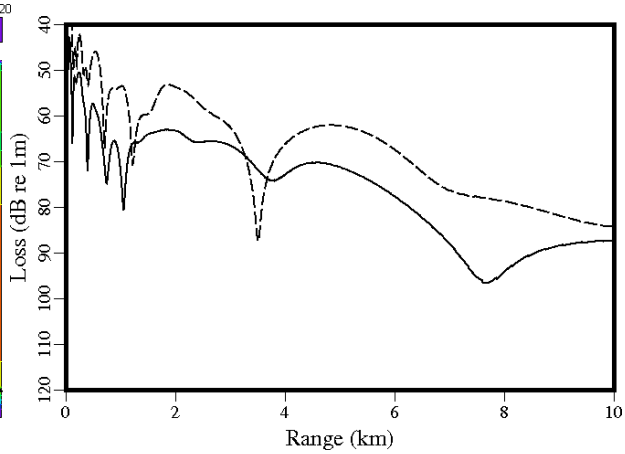


(a) The reference solution.

(b) The solution in the mapped domain.

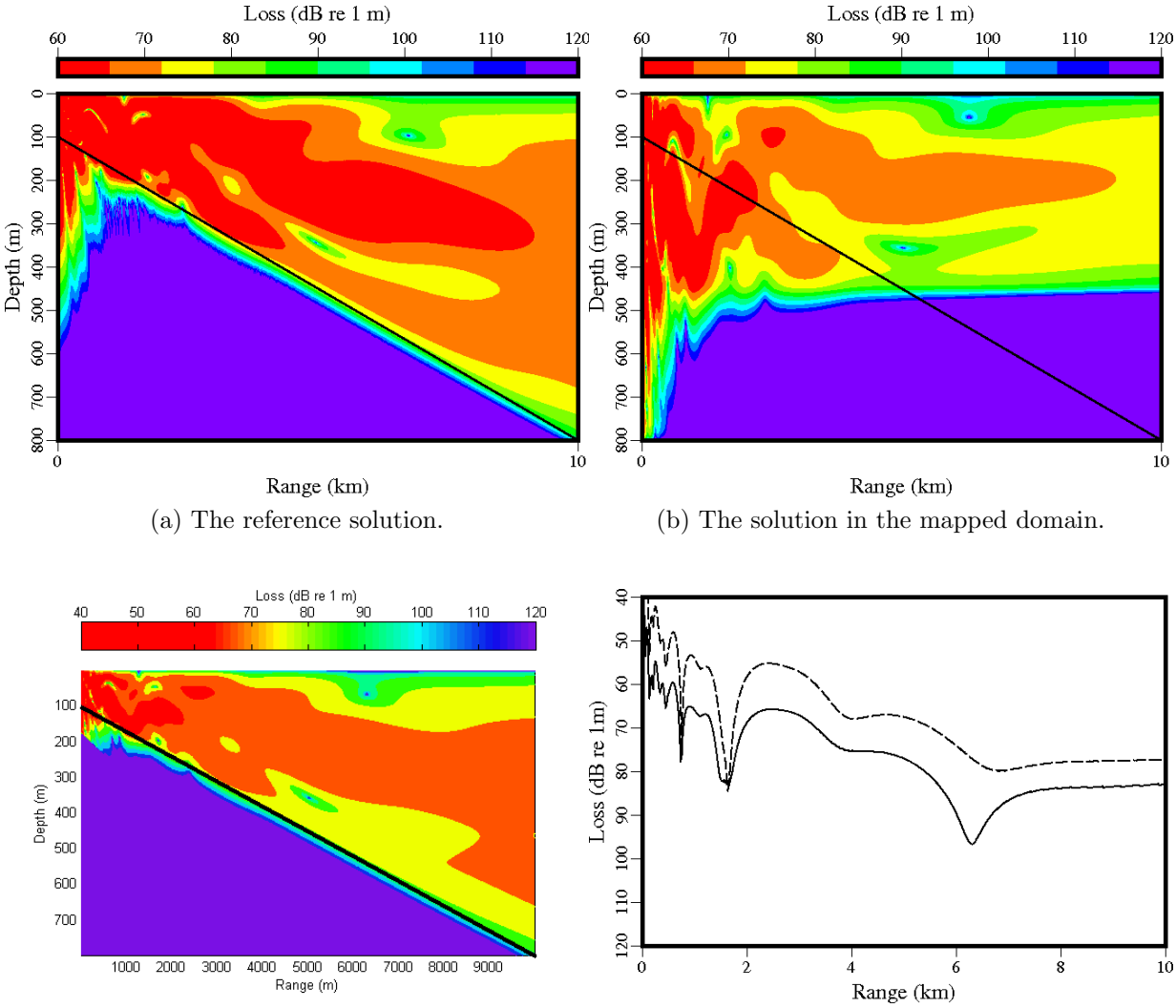


(c) The solution after mapping and domain adjustment have been applied.



(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.10: A $-3.1°$ bottom slope domain.



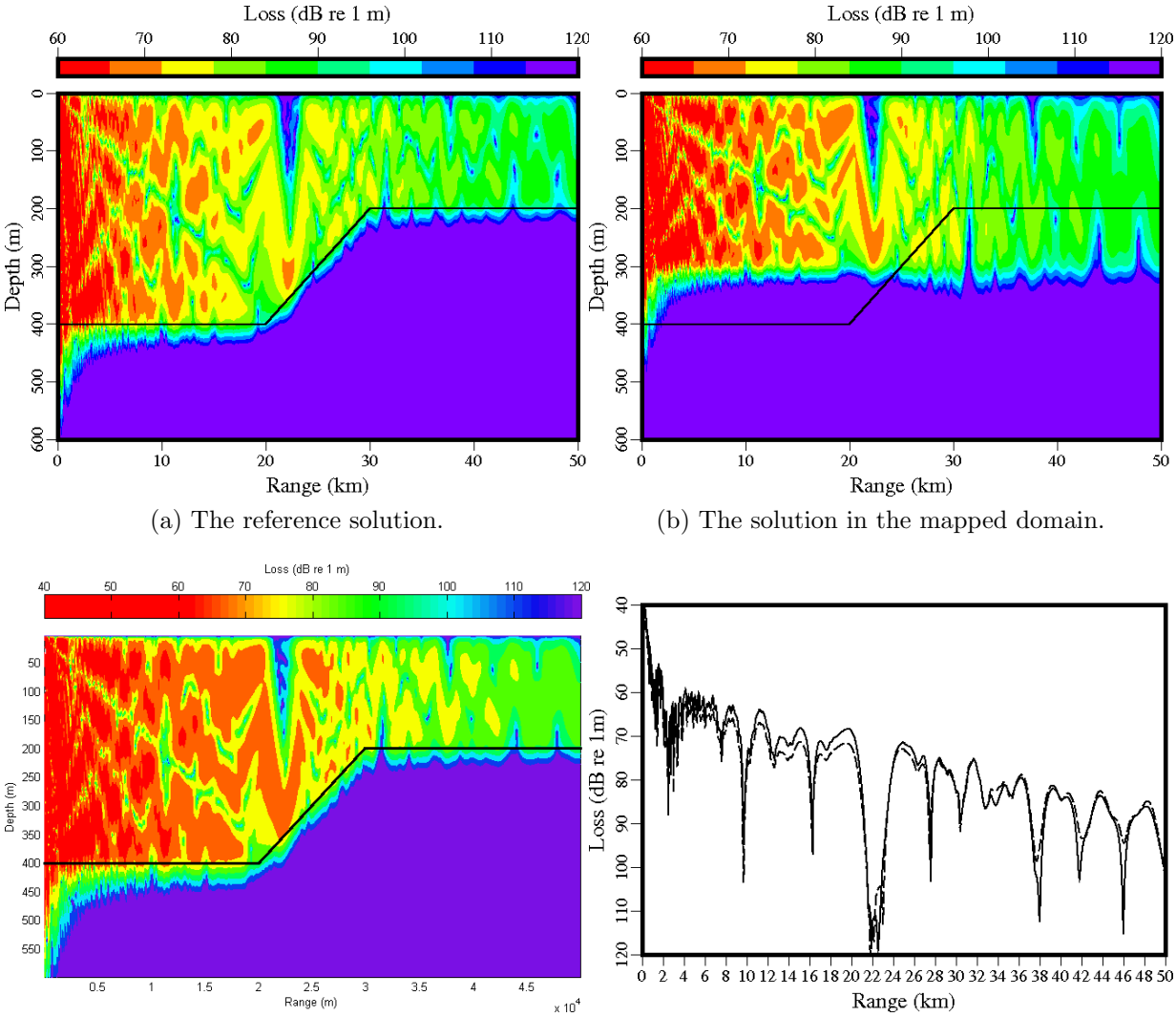
(a) The reference solution.

(b) The solution in the mapped domain.

(c) The solution after mapping and domain adjustment have been applied.

(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.11: A $-4.0°$ bottom slope domain.



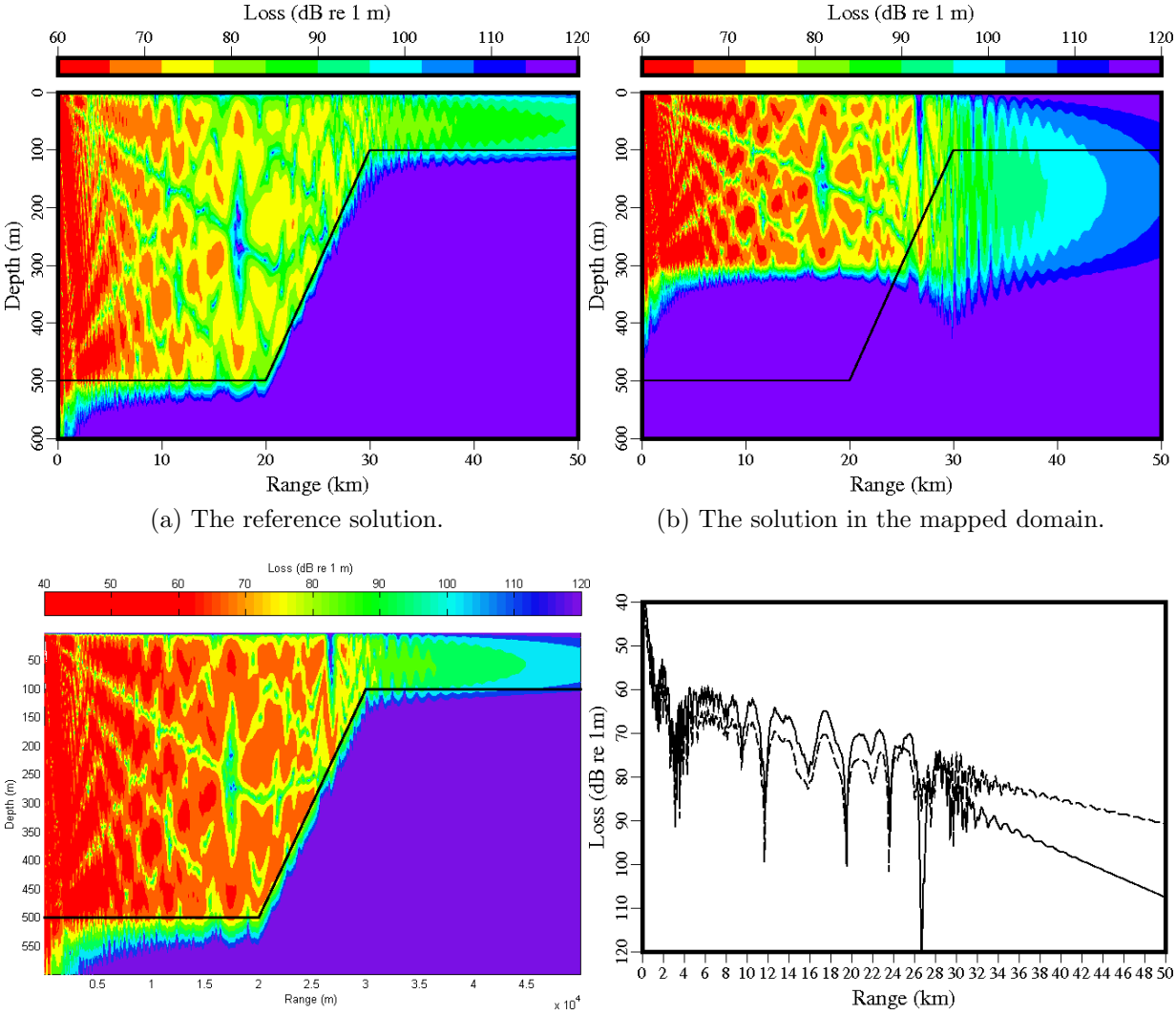
(a) The reference solution.

(b) The solution in the mapped domain.

(c) The solution after mapping and domain adjustment have been applied.

(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.12: A 1.1° variable bottom slope domain.



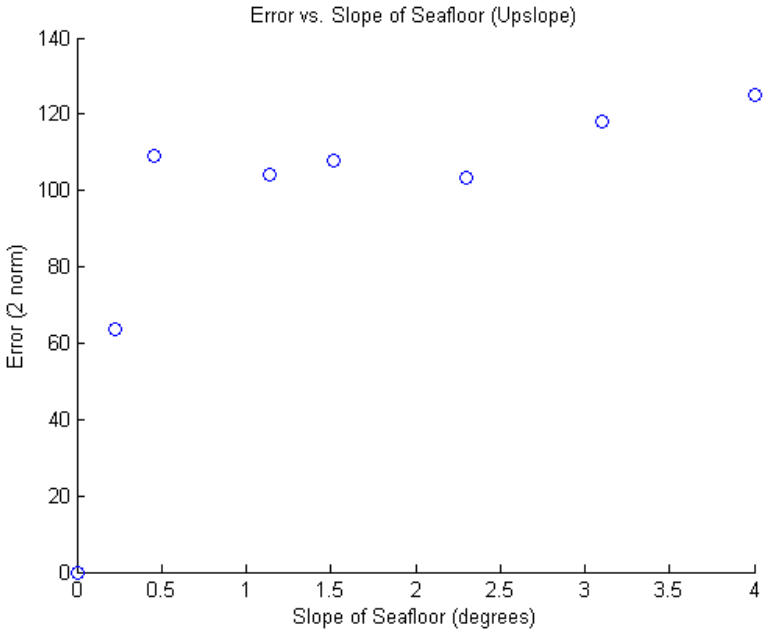
(a) The reference solution.

(b) The solution in the mapped domain.

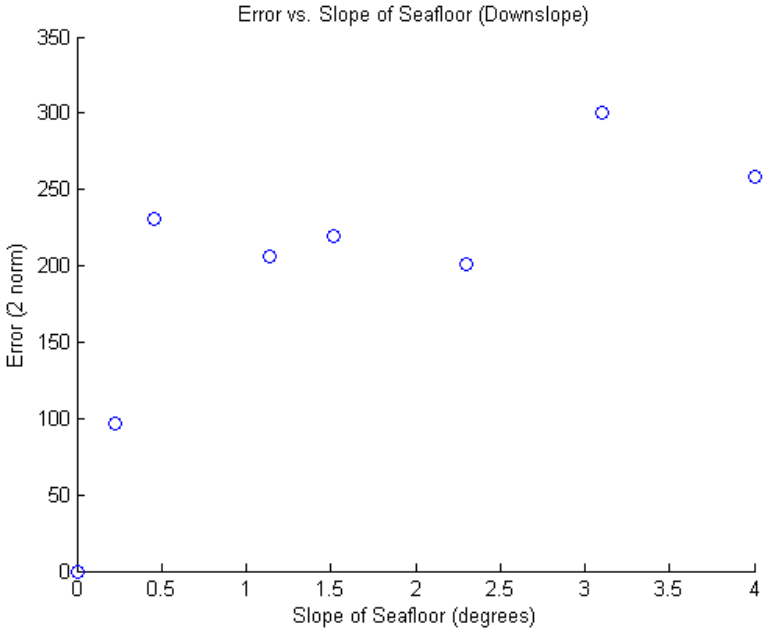
(c) The solution after mapping and domain adjustment have been applied.

(d) Transmission loss line plots at receiver depth 50 m. The solid curve represents the mapped solution and the dashed curve represents the reference solution (RAM)

Figure 4.13: A 2.3° variable bottom slope domain.



(a) The 2 norm of error calculated for upslope environments, plotted versus angle of slope.



(b) The 2 norm of error calculated for downslope environments, plotted versus angle of slope.

Figure 4.14: Error compared to slope for up and downslope environments.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

The parabolic equation method is able to provide efficient and accurate numerical techniques to model range-dependent sound propagation. Although the fluid medium parabolic equation has essentially been resolved, areas for improvement still exist. One key improvement to the solution of fluid equations is presented in this thesis, and relates to problems involving sloped seafloors.

A new coordinate mapping approach has been developed for solving problems involving sloping fluid-fluid interfaces, which has been an important area of research in the fluid parabolic equation. This new approach provides accurate PE solutions for environments involving range dependence and is benchmarked against quality results from a known solution.

The new coordinate mapping simplifies the acoustic environment. A sacrifice for a complex surface versus a complex bathymetry is no longer necessary, and this approach handles sloping interfaces accurately for shallow slopes. At each range where a change in slope occurs, the medium is mapped vertically so that a sloping interface becomes horizontal. To simplify the approach, the slope is assumed small and extra terms that arise in the wave equation because of the mapping are neglected.

The mapping solution was applied to range-dependent problems that involve relatively shallow slopes. Each solution is compared to a known valid solution to determine the validity of the solutions.

Several refinements and developments are required to make full use of this new mapping approach. Currently, only a few different, very simple environments have been tested. This solution must be expanded to work in environments with more complex bathymetry. The techniques used currently only concern environments where the slope of the floor is constant. Further work could be performed in expanding the solution to handle these more complex environments.

A resolution for the small errors in the solutions for upslope problems, and the larger error in downslope problems should be investigated. Originally, the errors were thought to arise from neglecting the additional terms in the wave equation, but evidence has been presented to the contrary.

A complete mapping correction could be derived to improve the accuracy of the coordinate mapping approach for range-dependent problems. A leading order correction is presented in (Collins & Dacol, 2000). However, this correction is based on approximations (approximately introducing equal but opposite bends each time the slope changes) and lacks an adequate analytical derivation. A complete correction will make this approach more applicable to various problems.

Another avenue for further work concerns the fact that the solution requires the use of various tools and manual modifications to obtain the final solution. An obvious course for future work is in the autonomization and simplification of the process of using this new mapping technique for range-dependent problems. Currently, the first solution is generated in the mapped domain, which then must be modified manually to a suitable input for another solution to map back into the initial domain. This solution would benefit greatly from simplifying the method used to obtain solutions.

Furthermore, after these modifications have been implemented, convergence testing needs to be carried out to determine for what environments the new mapping solution is valid. When performing convergence tests, the slope of the seafloor in particular needs to be considered to determine which slope angles in particular the mapped PE can handle.

An interesting consequence of the mapping solution to consider is multiple adjustments of the domain. The mapping forces a horizontal seafloor, but additionally, the corresponding parameters of the water column must also be adjusted to fit the new domain. It is likely that this adjustment has a large impact on the simplicity of the solution, because the interfaces between sound profiles must be adjusted as well.

Although this new mapping solution is reasonable in design (allowing for simpler interfaces to apply boundary conditions to) the additional computational cost of this solution in providing these simplifications should be considered. One cost is in determining the maximum depth at a given range step. Even though the depth is a parameter at the users control, the depth at a particular range step is linearly interpolated between depth values provided by the user. With these max depth values, another cost comes from applying the mapping to all values affected, which might result in stair-stepping. In essence, the benefits of the new method must outweigh the computational drawbacks this new mapping provides to see sustained use in the field of underwater acoustics.

REFERENCES CITED

Claerbout, J.F. 1976. *Fundamentals of Geophysical Data Processing.* McGraw Hill.

Collins, M. D. 1989. Applications and time-domain solution of higher-order parabolic equations in underwater acoustics. *"Journal of the Acoustical Society of America"*, **86**(3), 1097–1102.

Collins, M. D. Cederberg R. J., & King, D. B. 1995. Comparison of algorithms for solving parabolic wave equations. *Journal of the Acoustical Society of America*, **100**(1), 178–182.

Collins, M. D. Coury A., & Siegmann, W. L. 1995. Beach acoustics. *Journal of the Acoustical Society of America*, **97**(5), 2767–2770.

Collins, Michael D. 1993. A split-step Padé solution for the parabolic equation method. *Journal of the Acoustical Society of America*, **93**(4), 1736–1742.

Collins, Michael D. 1995. *User's Guide for RAM Versions 1.0 and 1.0p.*

Collins, Michael D., & Dacol, Dalcio K. 2000. A mapping approach for handling sloping interfaces. *Journal of the Acoustical Society of America*, **107**(4), 1357–1368.

Collins, Michael D., & Evans, Richard B. 1992. A two-way parabolic equation for acoustic backscattering in the ocean. *The Journal of the Acoustical Society of America*, **91**(3), 1357–1368.

Collins, Michael D., & Westwood, Evan K. 1991. A higher-order energy-conserving parabolic equation for range-dependent ocean depth, sound speed, and density. *Journal of the Acoustical Society of America*, **89**(3), 1068–1075.

DeSanto, John A. 1977. Relation between the solutions of the Helmhotz and parabolic equations for sound propagation. *The Journal of the Acoustical Society of America*, **62**(2), 295–297.

Jensen, F. B., Nielsen, P. L., Zampolli, M., Collins, M.D., & Siegmann, W.L. Benchmark scenarios for range-dependent seismo-acoustic models.

Küsel, Elizabeth Thorp. 2005. *New Parabolic Equation Solutions for High Frequency and Elastic Media Problems.* Ph.D. thesis, Rensselaer Polytechnic Institute.

Outing, Donald A. 2004. *Parabolic Equation Methods for Range Dependent Layered Elastic Media.* Ph.D. thesis, Rensselaer Polytechnic Institute.

Porter, M.B., Jensen, F. B., & Ferla, C.M. 1991. The problem of energy conservation in one-way models. *Journal of the Acoustical Society of America*, **89**(3), 1058–1067.

Tappert, F. D. 1977. The Parabolic Approximation Method. *Wave Propagation and Underwater Acoustics*, **70**, 224–287.

Tappert, F. D., Smith, K. B., & Wolfson, M. A. 2000. Analysis of the Split-Step Fourier Algorithm for the Solution of Parabolic Wave Equations. *Math. Model. Sci. Comp.*